

《偏微分方程》作业

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1 第 1 次作业

1. 设 $u(x_1, \dots, x_n) = f(r)$ (其中 $r = \sqrt{x_1^2 + \dots + x_n^2}$) 是 n 维调和函数 (即满足方程 $\frac{\partial^2 u}{\partial x_1^2} + \dots + \frac{\partial^2 u}{\partial x_n^2} = 0$), 试证明

$$f(r) = c_1 + \frac{c_2}{r^{n-2}} \quad (n \geq 2),$$

$$f(r) = c_1 + c_2 \ln \frac{1}{r} \quad (n = 2),$$

其中 c_1, c_2 为任意常数.

证. 直接计算可得 $u_{x_i} = f'(r) \frac{\partial r}{\partial x_i} = f'(r) \frac{x_i}{r}$, $u_{x_i x_i} = \frac{\partial u_{x_i}}{\partial x_i} = f''(r) \left(\frac{x_i}{r} \right)^2 + f'(r) \left(\frac{1}{r} - \frac{x_i}{r^2} \cdot \frac{x_i}{r} \right)$.
故

$$\Delta u = \sum_{i=1}^n u_{x_i x_i} = f''(r) + \frac{n-1}{r} f'(r).$$

由 $\Delta u = 0$ 知 $f''(r) + \frac{n-1}{r} f'(r) = 0$. 若 $f'(r) \neq 0$, 则有 $\frac{f''(r)}{f'(r)} = \frac{1-n}{r}$, 即

$$(\ln |f'(r)|)' = \frac{1-n}{r} = (1-n) \ln r'.$$

因此 $\ln |f'(r)| = (1-n) \ln r + \ln \tilde{c}_2 = \ln \frac{\tilde{c}_2}{r^{n-1}}$. 从而有 $f'(r) = \frac{\tilde{c}_2}{r^{n-1}}$. 故

$$f(r) = \begin{cases} c_1 + c_2 \ln \frac{1}{r}, & n = 2, \\ c_1 + \frac{c_2}{r^{n-2}}, & n \geq 3. \end{cases}$$

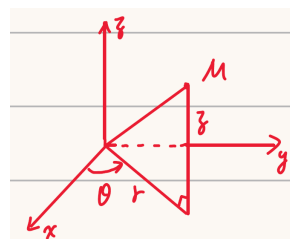
□

3. 证明: 拉普拉斯算子在柱坐标 (r, θ, z) 下可以写成

$$\Delta u = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{\partial^2 u}{\partial z^2}.$$

证. 柱坐标与直角坐标满足如下关系

$$\begin{cases} x = r \cos \theta, \\ y = r \sin \theta, \\ z = z. \end{cases}$$



于是有

$$\begin{cases} dx = d(r \cos \theta) = \cos \theta dr - r \sin \theta d\theta, \\ dy = d(r \sin \theta) = \sin \theta dr + r \cos \theta d\theta, \\ dz = dz. \end{cases}$$

从而可得

$$\begin{cases} dr = \cos \theta dx + \sin \theta dy, \\ d\theta = -\frac{1}{r} \sin \theta dx + \frac{1}{r} \cos \theta dy, \\ dz = dz, \end{cases}$$

故

$$\begin{aligned} \frac{\partial r}{\partial x} &= \cos \theta, \quad \frac{\partial r}{\partial y} = \sin \theta, \quad \frac{\partial \theta}{\partial x} = -\frac{1}{r} \sin \theta, \quad \frac{\partial \theta}{\partial y} = \frac{1}{r} \cos \theta, \\ \frac{\partial^2 r}{\partial x^2} &= -\sin \theta \frac{\partial \theta}{\partial x} = \frac{1}{r} \sin^2 \theta, \quad \frac{\partial^2 r}{\partial y^2} = \cos \theta \frac{\partial \theta}{\partial y} = \frac{1}{r} \cos^2 \theta, \\ \frac{\partial^2 \theta}{\partial x^2} &= \frac{1}{r^2} \sin \theta \frac{\partial r}{\partial x} - \frac{1}{r} \cos \theta \frac{\partial \theta}{\partial x} = \frac{2}{r^2} \sin \theta \cos \theta, \\ \frac{\partial^2 \theta}{\partial y^2} &= -\frac{1}{r^2} \cos \theta \frac{\partial r}{\partial y} - \frac{1}{r} \sin \theta \frac{\partial \theta}{\partial y} = -\frac{2}{r^2} \sin \theta \cos \theta. \end{aligned}$$

由

$$u_x = \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial x}$$

可得

$$\begin{aligned} u_{xx} &= \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial r} \right) \frac{\partial r}{\partial x} + \frac{\partial u}{\partial r} \frac{\partial^2 r}{\partial x^2} + \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial \theta} \right) \frac{\partial \theta}{\partial x} + \frac{\partial u}{\partial \theta} \frac{\partial^2 \theta}{\partial x^2} \\ &= \frac{\partial^2 u}{\partial r^2} \left(\frac{\partial r}{\partial x} \right)^2 + 2 \frac{\partial^2 u}{\partial \theta \partial r} \frac{\partial \theta}{\partial x} \frac{\partial r}{\partial x} + \frac{\partial^2 u}{\partial \theta^2} \left(\frac{\partial \theta}{\partial x} \right)^2 + \frac{\partial u}{\partial r} \frac{\partial^2 r}{\partial x^2} + \frac{\partial u}{\partial \theta} \frac{\partial^2 \theta}{\partial x^2}. \end{aligned}$$

同理由

$$u_y = \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial y}$$

可得

$$u_{yy} = \frac{\partial^2 u}{\partial r^2} \left(\frac{\partial r}{\partial y} \right)^2 + 2 \frac{\partial^2 u}{\partial \theta \partial r} \frac{\partial \theta}{\partial y} \frac{\partial r}{\partial y} + \frac{\partial^2 u}{\partial \theta^2} \left(\frac{\partial \theta}{\partial y} \right)^2 + \frac{\partial u}{\partial r} \frac{\partial^2 r}{\partial y^2} + \frac{\partial u}{\partial \theta} \frac{\partial^2 \theta}{\partial y^2}.$$

故

$$\begin{aligned} \Delta u &= u_{xx} + u_{yy} + u_{zz} \\ &= \frac{\partial^2 u}{\partial r^2} \left[\left(\frac{\partial r}{\partial x} \right)^2 + \left(\frac{\partial r}{\partial y} \right)^2 \right] + 2 \frac{\partial^2 u}{\partial \theta \partial r} \left(\frac{\partial \theta}{\partial x} \frac{\partial r}{\partial x} + \frac{\partial \theta}{\partial y} \frac{\partial r}{\partial y} \right) \\ &\quad + \frac{\partial^2 u}{\partial \theta^2} \left[\left(\frac{\partial \theta}{\partial x} \right)^2 + \left(\frac{\partial \theta}{\partial y} \right)^2 \right] + \frac{\partial u}{\partial r} \left(\frac{\partial^2 r}{\partial x^2} + \frac{\partial^2 r}{\partial y^2} \right) + \frac{\partial u}{\partial \theta} \left(\frac{\partial^2 \theta}{\partial x^2} + \frac{\partial^2 \theta}{\partial y^2} \right) + u_{zz} \\ &= \frac{\partial^2 u}{\partial r^2} (\cos^2 \theta + \sin^2 \theta) + 2 \frac{\partial^2 u}{\partial \theta \partial r} \frac{1}{r} (-\sin \theta \cos \theta + \sin \theta \cos \theta) + \frac{\partial^2 u}{\partial \theta^2} \frac{1}{r} (\sin^2 \theta + \cos^2 \theta) \\ &\quad + \frac{\partial u}{\partial r} \frac{1}{r^2} (\sin^2 \theta + \cos^2 \theta) + \frac{\partial u}{\partial \theta} \frac{2}{r^2} (\sin \theta \cos \theta - \sin \theta \cos \theta) + u_{zz} \\ &= \frac{\partial^2 u}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial z^2} \\ &= \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{\partial^2 u}{\partial z^2}. \end{aligned}$$

□

注 1.1. 上面过程的推导是将直角坐标系下的 Δ 转化为其在极坐标系下的表达式. 在不知道该表达式时, 上述过程是必要的. 但是若如题中已知极坐标下的表达式, 那么我们可以直接从该表达式出发, 也就是计算 $u_r, u_{rr}, u_\theta, u_{\theta\theta}$, 然后验证其即为直角坐标系下的 Δ . 这样计算量会更小.

4. 证明下列函数都是调和函数:

- (1) $ax + by + c$ (a, b, c 为常数);
- (2) $x^2 - y^2$ 和 $2xy$;
- (3) $x^3 - 3xy^2$ 和 $3x^2y - y^3$;
- (4) $\text{sh } ny \sin nx, \text{sh } ny \cos nx, \text{ch } ny \sin nx$ 和 $\text{ch } ny \cos nx$ (n 为常数);
- (5) $\text{sh } x(\text{ch } x + \cos y)^{-1}$ 和 $\sin y(\text{ch } x + \cos y)^{-1}$.

证. 只需证明 Laplace 算子作用之后为 0 即可.

(1) 记 $u(x, y) = ax + by + c$. $u_x = a, u_{xx} = 0, u_y = b, u_{yy} = 0$. 故 $\Delta u = u_{xx} + u_{yy} = 0$.

(2) 记 $u(x, y) = x^2 - y^2, v(x, y) = 2xy$.

$u_x = 2x, u_{xx} = 2, u_y = -2y, u_{yy} = -2$. 故 $\Delta u = 2 - 2 = 0$.

$v_x = 2y, v_{xx} = 0, v_y = 2x, v_{yy} = 0$. 故 $\Delta v = u_{xx} + u_{yy} = 0$.

(3) 记 $u(x, y) = x^3 - 3xy^2, v(x, y) = 3x^2y - y^3$.

$u_x = 3x^2 - 3y^2, u_{xx} = 6x, u_y = -6xy, u_{yy} = -6x$. 故 $\Delta u = 6x - 6x = 0$.

$v_x = 6xy, v_{xx} = 6y, v_y = 3x^2 - 3y^2, v_{yy} = -6y$. 故 $\Delta v = 6y - 6y = 0$.

(4) 记 $u(x, y) = \text{sh } ny \sin nx, v(x, y) = \text{sh } ny \cos nx, w(x, y) = \text{ch } ny \sin nx, \varphi(x, y) = \text{ch } ny \cos nx$.

由 $\text{sh } x = \frac{e^x - e^{-x}}{2}, \text{ch } x = \frac{e^x + e^{-x}}{2}$ 知 $(\text{sh } x)' = \text{ch } x, (\text{ch } x)' = \text{sh } x, (\text{sh } x)'' = \text{sh } x, (\text{ch } x)'' = \text{ch } x$. 又 $(\sin x)'' = -\sin x, (\cos x)'' = -\cos x$, 故

$$\Delta u = -n^2 \text{sh } ny \sin nx + n^2 \text{sh } ny \sin nx = 0,$$

$$\Delta v = -n^2 \text{sh } ny \cos nx + n^2 \text{sh } ny \cos nx = 0,$$

$$\Delta w = -n^2 \text{ch } ny \sin nx + n^2 \text{ch } ny \sin nx = 0,$$

$$\Delta \varphi = -n^2 \text{ch } ny \cos nx + n^2 \text{ch } ny \cos nx = 0.$$

(5) 记 $u(x, y) = \text{sh } x(\text{ch } x + \cos y)^{-1}, v(x, y) = \sin y(\text{ch } x + \cos y)^{-1}$.

由

$$\begin{aligned} u_x &= \frac{\text{ch } x(\text{ch } x + \cos y) - \text{sh } x \text{sh } x}{(\text{ch } x + \cos y)^2} = \frac{\text{ch}^2 x - \text{sh}^2 x + \text{ch } x \cos y}{(\text{ch } x + \cos y)^2}, \\ u_{xx} &= \frac{(2 \text{ch } x \text{sh } x - 2 \text{sh } x \text{ch } x + \text{sh } x \cos y)(\text{ch } x + \cos y)^2 - (\text{ch}^2 x - \text{sh}^2 x + \text{ch } x \cos y) \cdot 2(\text{ch } x + \cos y) \text{sh } x}{(\text{ch } x + \cos y)^4} \\ &= \frac{\text{sh } x \cos y(\text{ch } x + \cos y) - 2 \text{sh } x(1 + \text{ch } x \cos y)}{(\text{ch } x + \cos y)^3} = \frac{\text{sh } x \cos^2 y - \text{sh } x \text{ch } x \cos y - 2 \text{sh } x}{(\text{ch } x + \cos y)^3} \end{aligned}$$

和

$$\begin{aligned}
u_y &= \frac{\operatorname{sh} x \sin y}{(\operatorname{ch} x + \cos y)^2}, \\
u_{yy} &= \frac{\operatorname{sh} x \cos y (\operatorname{ch} x + \cos y)^2 + 2 \operatorname{sh} x \sin y (\operatorname{ch} x + \cos y) \sin y}{(\operatorname{ch} x + \cos y)^4} \\
&= \frac{\operatorname{sh} x \cos y (\operatorname{ch} x + \cos y) + 2 \operatorname{sh} x \sin^2 y}{(\operatorname{ch} x + \cos y)^3}, \\
&= \frac{\operatorname{sh} x (\cos^2 y + 2 \sin^2 y) + \operatorname{sh} x \operatorname{ch} x \cos y}{(\operatorname{ch} x + \cos y)^3}
\end{aligned}$$

可得

$$\Delta u = u_{xx} + u_{yy} = \frac{\operatorname{sh} x (2 \cos^2 y + 2 \sin^2 y) - 2 \operatorname{sh} x}{(\operatorname{ch} x + \cos y)^3} = 0.$$

由

$$\begin{aligned}
v_x &= -\frac{\operatorname{sh} x \sin y}{(\operatorname{ch} x + \cos y)^2}, \\
v_{xx} &= -\frac{\operatorname{ch} x \sin y (\operatorname{ch} x + \cos y)^2 - \operatorname{sh} x \sin y \cdot 2(\operatorname{ch} x + \cos y) \operatorname{sh} x}{(\operatorname{ch} x + \cos y)^4} \\
&= \frac{-\operatorname{ch}^2 x \sin y - \operatorname{ch} x \sin y \cos y + 2 \operatorname{sh}^2 x \sin y}{(\operatorname{ch} x + \cos y)^3}
\end{aligned}$$

和

$$\begin{aligned}
v_y &= \frac{\cos y (\operatorname{ch} x + \cos y) + \sin^2 y}{(\operatorname{ch} x + \cos y)^2} = \frac{\operatorname{ch} x \cos y + 1}{(\operatorname{ch} x + \cos y)^2}, \\
v_{yy} &= \frac{-\operatorname{ch} x \sin y (\operatorname{ch} x + \cos y)^2 + (\operatorname{ch} x \cos y + 1) \cdot 2(\operatorname{ch} x + \cos y) \sin y}{(\operatorname{ch} x + \cos y)^4} \\
&= \frac{-\operatorname{ch}^2 x \sin y + \operatorname{ch} x \sin y \cos y + 2 \sin y}{(\operatorname{ch} x + \cos y)^3}
\end{aligned}$$

可得

$$\Delta u = u_{xx} + u_{yy} = \frac{(-2 \operatorname{ch}^2 x + 2 \operatorname{sh}^2 x) \sin y + 2 \sin y}{(\operatorname{ch} x + \cos y)^3} = 0.$$

□

补充题 1. 利用 Gauss–Green 定理证明二维 Green 公式:

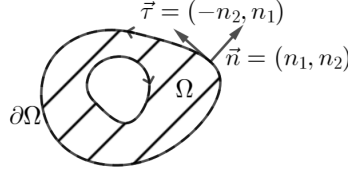
$$\iint_{\Omega} (Q_x - P_y) \, dx \, dy = \oint_{\partial \Omega} P \, dx + Q \, dy,$$

其中 $P(x, y), Q(x, y) : \Omega \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$, Ω 是一开区域, $\partial \Omega$ 为右手方向.

证. 由 Gauss–Green 定理知

$$\iint_{\Omega} (Q_x - P_y) \, dx \, dy = \int_{\partial \Omega} Q \cdot n_1 \, ds - \int_{\partial \Omega} P \cdot n_2 \, ds,$$

其中 $\vec{n} = (n_1, n_2)$ 为 $\partial \Omega$ 上的单位外法向. 若考虑 $\partial \Omega$ 在右手方向上的切向 $\vec{\tau}$, 则有 $\vec{\tau} = (-n_2, n_1)$.



如图所示, 知 $dx = -n_2 ds$, $dy = n_1 ds$, 故

$$\iint_{\Omega} (Q_x - P_y) dx dy = \oint_{\partial\Omega} P dx + Q dy.$$

□

补充题 2. 利用 Gauss–Green 定理证明三维 Green 公式 (Gauss 公式)

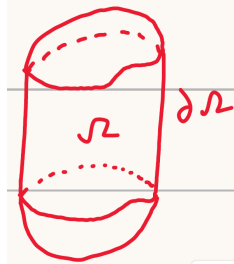
$$\iiint_{\Omega} (P_x + Q_y + R_z) dx dy dz = \iiint_{\partial\Omega} P dy dz + Q dz dx + R dx dy$$

其中 $P(x, y, z), Q(x, y, z), R(x, y, z) : \mathbb{R}^3 \rightarrow \mathbb{R}$, Ω 是一开区域, $\partial\Omega$ 取外侧.

证. 由 Gauss–Green 定理知

$$\iiint_{\Omega} (P_x + Q_y + R_z) dx dy dz = \iint_{\partial\Omega} (P n_1 + Q n_2 + R n_3) dS,$$

其中 $\vec{n}$ 为 $\partial\Omega$ 上的单位外法向. 因此 $\vec{n}$ 是 $\partial\Omega$ 上切平面的法向.



考虑 $\partial\Omega$ 上某点的面微元 dS . 记该点上切平面与 xOy 平面的夹角为 α , 于是 dS 在 xOy 上的投影为 $\cos \alpha dS$. 由于 α 又是切平面法向与 xOy 平面法向的夹角, 故 $\cos \alpha = \vec{n} \cdot (0, 0, 1) = n_3$. 故 dS 在 xOy 上的投影为 $n_3 dS$. 同理 dS 在 xOz, yOz 上的投影分别是 $n_2 dS, n_1 dS$. 故

$$\iint_{\partial\Omega} P n_1 dS + \iint_{\partial\Omega} Q n_2 dS + \iint_{\partial\Omega} R n_3 dS = \iiint_{\partial\Omega} P dy dz + Q dz dx + R dx dy.$$

从而有

$$\iiint_{\Omega} (P_x + Q_y + R_z) dx dy dz = \iiint_{\partial\Omega} P dy dz + Q dz dx + R dx dy.$$

□

2 第 2 次作业

12. 设

$$J(v) = \iiint_{\Omega} \frac{1}{2} \left[\left(\frac{\partial v}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial v}{\partial z} \right)^2 \right] dx dy dz + \iint_{\Gamma} \left(\frac{1}{2} \sigma v^2 - gv \right) dS,$$

其中 $\sigma > 0$ 为一给定常数. 变分问题的提法为: 求 $u \in V$, 使

$$J(u) = \min_{v \in V} J(v),$$

其中 $V = C^2(\Omega) \cap C^1(\bar{\Omega})$. 试导出与此变分问题等价的边值问题, 并证明它们的等价性.

解. 对 $\forall v \in V, \lambda \in \mathbb{R}$, 有

$$\begin{aligned} J(u + \lambda v) &= \iiint_{\Omega} \frac{1}{2} |\nabla u + \lambda \nabla v|^2 dx dy dz + \iint_{\partial\Omega} \left[\frac{1}{2} \sigma (u + \lambda v)^2 - g \cdot (u + \lambda v) \right] dS \\ &= \iiint_{\Omega} \frac{1}{2} |\nabla u|^2 dx dy dz + \iint_{\Omega} \left(\frac{1}{2} \sigma u^2 - gu \right) dS \\ &\quad + \lambda \left[\iiint_{\Omega} \nabla u \cdot \nabla v dx dy dz + \iint_{\partial\Omega} (\sigma uv - gv) dS \right] \\ &\quad + \lambda^2 \left(\iiint_{\Omega} \frac{1}{2} |\nabla v|^2 dx dy dz + \iint_{\partial\Omega} \frac{1}{2} \sigma v^2 dS \right). \end{aligned}$$

由于 $J(u + \lambda v)$ 在 $\lambda = 0$ 处取到最小值, 故

$$0 = \frac{dJ(u + \lambda v)}{d\lambda} \Big|_{\lambda=0} = \iiint_{\Omega} \nabla u \cdot \nabla v dx dy dz + \iint_{\partial\Omega} (\sigma uv - gv) dS.$$

由 Green 公式知对 $\forall v \in V$ 有

$$- \iiint_{\Omega} v \Delta u dx dy dz + \iint_{\partial\Omega} \left(\frac{\partial u}{\partial \vec{n}} + \sigma u - g \right) v dS = 0.$$

下先证 $\Delta u = 0$ 在 Ω 上恒成立. 反证而设存在 $M_0 \in \Omega$, 使得 $\Delta u(M_0) \neq 0$. 不妨设 $-\Delta u(M_0) > 0$. 由 $u \in C^2(\Omega)$ 知 $\Delta u \in C(\Omega)$, 故存在 M_0 的邻域 $U_1 \subset \Omega$, 使得 $-\Delta u|_{U_1} > 0$. 取 M_0 的邻域 $U_2 \subsetneq U_1$, 并取 $v \in C^2(\Omega) \cap C^1(\bar{\Omega})$ 使得 $v|_{U_1} \geq 0, v|_{U_2} > 0, v|_{\Omega \setminus U_1} = 0$. 于是有

$$- \iiint_{\Omega} v \Delta u dx dy dz + \iint_{\partial\Omega} \left(\frac{\partial u}{\partial \vec{n}} + \sigma u - g \right) v dS = - \iiint_{U_1} v \Delta u dx dy dz \geq - \iiint_{U_2} v \Delta u dx dy dz > 0.$$

矛盾, 故 $\Delta u = 0$ 在 Ω 上恒成立. 于是有 $\iint_{\partial\Omega} \left(\frac{\partial u}{\partial \vec{n}} + \sigma u - g \right) v dS = 0$. 同理, 利用 $v \in C^1(\bar{\Omega})$ 的任意性知 $\left(\frac{\partial u}{\partial \vec{n}} + \sigma u - g \right) \Big|_{\partial\Omega} = 0$. 综上, 有

$$\begin{cases} \Delta u = 0, & \text{在 } \Omega \text{ 上,} \\ \left(\frac{\partial u}{\partial \vec{n}} + \sigma u \right) \Big|_{\partial\Omega} = g. \end{cases} \quad (2.1)$$

此即为由题设中变分问题导出的边值问题.

下证该边值问题的解即为题设中变分问题的解, 这就证明了等价性.

设 $u \in V$ 是边值问题 (2.1) 的解. 对 $\forall v \in V$, 记 $w = v - u$. 则有

$$\begin{aligned} J(v) &= J(u + w) = J(u) + \iiint_{\Omega} \nabla u \cdot \nabla w dx dy dz + \iint_{\partial\Omega} (\sigma u - g) w dS \\ &\quad + \iiint_{\Omega} \frac{1}{2} |\nabla w|^2 dx dy dz + \iint_{\partial\Omega} \frac{1}{2} \sigma w^2 dS \\ &\geq J(u) + \iiint_{\Omega} \nabla u \cdot \nabla w dx dy dz + \iint_{\partial\Omega} (\sigma u - g) w dS. \end{aligned}$$

由 (2.1) 以及 Green 公式知

$$\begin{aligned} &\iiint_{\Omega} \nabla u \cdot \nabla w dx dy dz + \iint_{\partial\Omega} (\sigma u - g) w dS \\ &= - \iiint_{\Omega} w \Delta u dx dy dz + \iint_{\partial\Omega} \left(\frac{\partial u}{\partial \vec{n}} + \sigma u - g \right) w dS \\ &= 0. \end{aligned}$$

故 $J(v) \geq J(u)$ 对 $\forall v \in V$ 成立, 即 $J(u) = \min_{v \in V} J(v)$. 证毕. $\square$

2. 若函数 $u(x, y)$ 是单位圆盘上的调和函数, 又它在单位圆周上的数值已知为 $u = \sin \theta$, 其中 θ 表示极角, 问函数 u 在原点之值等于多少?

解. 记单位圆盘为 Ω , 则由平均值公式知 $u(0) = \frac{1}{2\pi} \int_{\partial\Omega} u ds$. $\partial\Omega$ 的参数方程为 $x = \cos \theta, y = \sin \theta$. 于是有

$$ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{(\sin^2 \theta + \cos^2 \theta) (d\theta)^2} = d\theta.$$

故

$$u(0) = \frac{1}{2\pi} \int_{\partial\Omega} u ds = \frac{1}{2\pi} \int_0^{2\pi} \sin \theta d\theta = 0.$$

$\square$

3. 设 $u(M)$ 在 Ω 内调和, M_0 是 Ω 中的任意点. B_a 是以 M_0 为球心, a 为半径的球体, 其体积为 $|B_a|$, 证明: 成立

$$u(M_0) = \frac{1}{|B_a|} \iiint_{B_a} u dx dy dz$$

证. 记 B_r 为以 M_0 为球心 r 为半径的球, 则由 $u(M_0) = \frac{1}{4\pi r^2} \iint_{\partial B_r} u dS$ 知

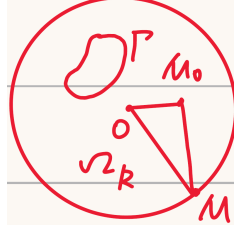
$$\iiint_{B_a} u dx dy dz = \int_0^a \left(\iint_{\partial B_r} u dS \right) dr = \int_0^a 4\pi r^2 u(M_0) dr = \frac{4}{3} \pi a^3 u(M_0).$$

结合 $|B_a| = \frac{4}{3} \pi a^3$ 得 $u(M_0) = \frac{1}{|B_a|} \iiint_{B_a} u dx dy dz$. $\square$

4. 证明: 当 $u(M)$ 在闭曲面 Γ 的外部调和, 并且在无穷远处成立

$$u(M) = O\left(\frac{1}{r_{OM}}\right), \quad \frac{\partial u}{\partial r} = O\left(\frac{1}{r_{OM}^2}\right) \quad (r_{OM} \rightarrow \infty),$$

而 M_0 是 Γ 外的任一点, 则公式 (2.6) 仍成立.



证. 记 $\Omega_R = \{M \in \mathbb{R}^3 : r_{OM} < R \text{ 且 } M \text{ 在 } \Gamma \text{ 的外部}\}$. 对 Γ 外任一点 M_0 , 对任意 $R > |r_{OM_0}|$, 有 $M_0 \in \Omega_R$. 于是有

$$\begin{aligned} u(M_0) &= -\frac{1}{4\pi} \iint_{\partial\Omega_R} \left[u(M) \frac{\partial}{\partial \vec{n}} \left(\frac{1}{r_{M_0M}} \right) - \frac{1}{r_{M_0M}} \frac{\partial u(M)}{\partial \vec{n}} \right] dS \\ &= -\frac{1}{4\pi} \iint_{\Gamma} \left[u(M) \frac{\partial}{\partial \vec{n}} \left(\frac{1}{r_{M_0M}} \right) - \frac{1}{r_{M_0M}} \frac{\partial u(M)}{\partial \vec{n}} \right] dS \\ &\quad - \frac{1}{4\pi} \iint_{r_{OM}=R} \left[u(M) \frac{\partial}{\partial \vec{n}} \left(\frac{1}{r_{M_0M}} \right) - \frac{1}{r_{M_0M}} \frac{\partial u(M)}{\partial \vec{n}} \right] dS. \end{aligned} \quad (2.2)$$

在球面 $r_{OM} = R$ 上, $\vec{n} = \frac{\overrightarrow{OM}}{r_{OM}}$, 该方向上点的球坐标只改变径向距离, 不改变极角与方位角. 故此时

$$\frac{\partial}{\partial \vec{n}} \left(\frac{1}{r_{M_0M}} \right) = -\frac{\overrightarrow{M_0M}}{r_{M_0M}^3} \cdot \frac{\overrightarrow{OM}}{r_{OM}} = -\frac{r_{OM}^2 - \overrightarrow{OM} \cdot \overrightarrow{OM_0}}{r_{M_0M}^3 r_{OM}}, \quad \frac{\partial u}{\partial \vec{n}} = \frac{\partial u}{\partial r}.$$

由 $r_{M_0M} \geq r_{OM} - r_{OM_0} > 0$ 知

$$\begin{aligned} &\left| \iint_{r_{OM}=R} \left[u(M) \frac{\partial}{\partial \vec{n}} \left(\frac{1}{r_{M_0M}} \right) - \frac{1}{r_{M_0M}} \frac{\partial u(M)}{\partial \vec{n}} \right] dS \right| \\ &\leq \iint_{r_{OM}=R} |u(M)| \frac{|r_{OM}^2 - \overrightarrow{OM} \cdot \overrightarrow{OM_0}|}{r_{M_0M}^3 r_{OM}} dS + \iint_{r_{OM}=R} \frac{1}{r_{M_0M}} \left| \frac{\partial u(M)}{\partial r} \right| dS \\ &\leq C \left(\iint_{r_{OM}=R} \frac{1}{r_{OM}} \frac{r_{OM} + r_{OM_0}}{r_{M_0M}^3} dS + \iint_{r_{OM}=R} \frac{1}{r_{M_0M}} \frac{1}{r_{OM}^2} dS \right) \\ &\leq C \left(\frac{R + r_{OM_0}}{R(R - r_{OM_0})^3} + \frac{1}{R - r_{OM_0}} \frac{1}{R^2} \right) \iint_{r_{OM}=R} 1 dS \\ &= \frac{4\pi CR}{(R - r_{OM_0})} \left(\frac{R + r_{OM_0}}{(R - r_{OM_0})^2} + \frac{1}{R} \right) \rightarrow 0, \quad \text{当 } R \rightarrow \infty, \end{aligned}$$

其中 C 为某一常数. 故 $\lim_{R \rightarrow +\infty} \iint_{r_{OM}=R} \left[u(M) \frac{\partial}{\partial \vec{n}} \left(\frac{1}{r_{M_0M}} \right) - \frac{1}{r_{M_0M}} \frac{\partial u(M)}{\partial \vec{n}} \right] dS = 0$. 在 (2.2) 中令 $R \rightarrow +\infty$, 则得

$$u(M_0) = -\frac{1}{4\pi} \iint_{\Gamma} \left[u(M) \frac{\partial}{\partial \vec{n}} \left(\frac{1}{r_{M_0M}} \right) - \frac{1}{r_{M_0M}} \frac{\partial u(M)}{\partial \vec{n}} \right] dS.$$

□

3 第 3 次作业

5. 证明调和方程狄利克雷外问题解的稳定性.

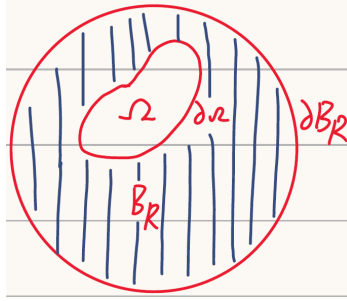
证. 设 u 与 u^* 分别满足

$$\begin{cases} -\Delta u = f & \text{在 } \mathbb{R}^3 \setminus \bar{\Omega} \text{ 上} \\ u|_{\partial\Omega} = g, \\ \lim_{r \rightarrow \infty} u(x, y, z) = 0, \end{cases} \quad \text{和} \quad \begin{cases} -\Delta u^* = f & \text{在 } \mathbb{R}^3 \setminus \bar{\Omega} \text{ 上} \\ u^*|_{\partial\Omega} = g^*, \\ \lim_{r \rightarrow \infty} u^*(x, y, z) = 0. \end{cases}$$

令 $v = u - u^*$, 则

$$\begin{cases} \Delta v = 0 & \text{在 } \mathbb{R}^3 \setminus \bar{\Omega} \text{ 上} \\ v|_{\partial\Omega} = g - g^*, \\ \lim_{r \rightarrow \infty} v(x, y, z) = 0. \end{cases}$$

记 $B_R = \{(x, y, z) \in \mathbb{R}^3 : \sqrt{x^2 + y^2 + z^2} < R\}$, $\Omega_R = B_R \setminus \bar{\Omega}$. 由 v 在 Ω_R 上调和知



$$\begin{aligned} \max_{\bar{\Omega}_R} v &= \max_{\partial\Omega \cup \partial B_R} v = \max \left\{ \max_{\partial\Omega} (g - g^*), \max_{\partial B_R} v \right\}, \\ \min_{\bar{\Omega}_R} v &= \min_{\partial\Omega \cup \partial B_R} v = \min \left\{ \min_{\partial\Omega} (g - g^*), \min_{\partial B_R} v \right\}. \end{aligned}$$

令 $R \rightarrow \infty$, 则

$$\begin{aligned} \sup_{\mathbb{R}^n \setminus \Omega} v &= \lim_{R \rightarrow \infty} \max_{\bar{\Omega}_R} v = \lim_{R \rightarrow \infty} \max \left\{ \max_{\partial R} (g - g^*), \max_{\partial B_R} v \right\} \\ &= \max \left\{ \max_{\partial R} (g - g^*), \lim_{R \rightarrow \infty} \max_{\partial B_R} v \right\} \\ &= \max \left\{ \max_{\partial\Omega} (g - g^*), 0 \right\} \\ &\leq \max_{\partial B_R} |g - g^*| \end{aligned}$$

且

$$\begin{aligned}
\inf_{\mathbb{R}^n \setminus \Omega} v &= \lim_{R \rightarrow \infty} \min_{\Omega_R} v = \lim_{R \rightarrow \infty} \min \left\{ \min_{\partial \Omega} (g - g^*), \min_{\partial B_R} v \right\} \\
&= \min \left\{ \min_{\partial \Omega} (g - g^*), \lim_{R \rightarrow \infty} \min_{\partial B_R} v \right\} \\
&= \min \left\{ \min_{\partial \Omega} (g - g^*), 0 \right\} \\
&\geq -\max_{\partial \Omega} |g - g^*|,
\end{aligned}$$

即 $-\max |g - g^*| \leq v(M) \leq \max_{\partial \Omega} |g - g^*|$, $\forall M \in \mathbb{R}^3 \setminus \bar{\Omega}$. 故 $\sup_{\mathbb{R}^3 \setminus \Omega} |u - u^*| \leq \max_{\partial \Omega} |g - g^*|$,

即 $\|u - u^*\|_{L^\infty(\mathbb{R}^3 \setminus \bar{\Omega})} \leq \|g - g^*\|_{L^\infty(\partial \Omega)}$. 因此调和方程狄利克雷外问题的解连续依赖于边界条件 g . $\square$

7. 证明第 6 题中讨论的椭圆型方程的第一边值问题解的唯一性与稳定性.

证. 设 u_1, u_2 是

$$\begin{cases} \sum_{i,j=1}^n a_{ij} \frac{\partial^2 u}{\partial x_i \partial x_j} + \sum_{i=1}^n b_i \frac{\partial u}{\partial x_i} + cu = f & \text{在 } \Omega \text{ 上} \\ u|_{\partial \Omega} = g \end{cases} \quad (3.1)$$

的解. 令 $w = u_1 - u_2$, 则

$$\begin{cases} \sum_{i,j=1}^n a_{ij} \frac{\partial^2 w}{\partial x_i \partial x_j} + \sum_{i=1}^n b_i \frac{\partial w}{\partial x_i} + cw = 0 & \text{在 } \Omega \text{ 上,} \\ w|_{\partial \Omega} = 0. \end{cases}$$

由题 6 知 w 不能在 Ω 的内部达到正的最大值或负的最小值, 故

$$\max_{\bar{\Omega}} |w| \leq \max_{\partial \Omega} |w| = 0.$$

从而在 $\bar{\Omega}$ 上 $w \equiv 0$, 即在 $\bar{\Omega}$ 上 $u_1 \equiv u_2$. 因此 (3.1) (1) 的解存在则必唯一.

设 u 是 (3.1) 的解, u^* 满足

$$\begin{cases} \sum_{i,j=1}^n a_{ij} \frac{\partial^2 u^*}{\partial x_i \partial x_j} + \sum_{i=1}^n b_i \frac{\partial u^*}{\partial x_i} + cu^* = f & \text{在 } \Omega \text{ 上,} \\ u^*|_{\partial \Omega} = g^* \end{cases}$$

令 $v = u - u^*$, 则

$$\begin{cases} \sum_{i,j=1}^n a_{ij} \frac{\partial^2 v}{\partial x_i \partial x_j} + \sum_{i=1}^n b_i \frac{\partial v}{\partial x_i} + cv = 0 & \text{在 } \Omega \text{ 上,} \\ v|_{\partial \Omega} = g - g^*. \end{cases}$$

同上可知 $\max_{\Omega} |v| \leq \max_{\partial \Omega} |v| = \max_{\partial \Omega} |g - g^*|$, 即

$$\|u - u^*\|_{L^\infty(\Omega)} \leq \|g - g^*\|_{L^\infty(\partial \Omega)}.$$

故 (3.1) 的解连续依赖于边界条件 g . $\square$

1. 证明格林函数的性质 3 及性质 5.

证. (1) 性质 3: $0 < G(M_1, M_0) < \frac{1}{4\pi r_{M_0 M}}, \forall M, M_0 \in \Omega, M \neq M_0$.

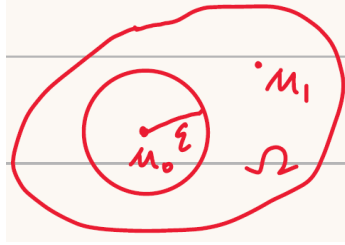
由 $g(\cdot, M_0)$ 满足

$$\begin{cases} \Delta g(\cdot, M_0) = 0 & \text{在 } \Omega \text{ 上} \\ g(\cdot, M_0)|_{\partial\Omega} = \frac{1}{4\pi r_{M_0 M}} \end{cases}$$

知 $g(M, M_0) > \min_{\partial\Omega} g(\cdot, M_0) > 0, \forall M \in \Omega$. 故 $G(M, M_0) = \frac{1}{4\pi r_{M_0 M}} - g(M, M_0) < \frac{1}{4\pi r_{M_0 M}}, \forall M \in \Omega$. 又 $\max_{\Omega} g(\cdot, M_0) < \max_{\partial\Omega} g(\cdot, M_0) < +\infty$, 故对 $\forall M_1 \in \Omega$, 可取 $\varepsilon > 0$ 足够小, 使得 $M_1 \in \Omega \setminus \overline{B(M_0, \varepsilon)}$ 且对 $\forall M \in \partial B(M_0, \varepsilon)$ 有

$$G(M, M_0) = \frac{1}{4\pi r_{M_0 M}} - g(M, M_0) > 0.$$

由 $G(\cdot, M_0)$ 在 $\Omega \setminus \overline{B(M_0, \varepsilon)}$ 上调和知



$$G(M_1, M_0) > \min_{\partial\Omega \cup \partial B(M_0, \varepsilon)} G(\cdot, M_0) = \min \left\{ 0, \min_{\partial B(M_0, \varepsilon)} G(M, M_0) \right\} = 0.$$

综上, 结合 $M_1 \in \Omega$ 的任意性知 $0 < G(M_1, M_0) < \frac{1}{4\pi r_{M_0 M}}, \forall M \in \Omega$.

$$(2) \text{ 性质 5: } \iint_{\partial\Omega} \frac{\partial G(M_1, M_0)}{\partial \vec{n}} dS = -1.$$

由 $g(\cdot, M_0)$ 在 Ω 上调和知 $\iint_{\partial\Omega} \frac{\partial g(M, M_0)}{\partial \vec{n}} dS = 0$. 故

$$\iint_{\partial\Omega} \frac{\partial G(M, M_0)}{\partial \vec{n}} dS = \iint_{\partial\Omega} \frac{\partial}{\partial \vec{n}} \left(\frac{1}{4\pi r_{M_0 M}} \right) dS.$$

利用 (1) 中的 $B(M_0, \varepsilon)$. 由 $\frac{1}{4\pi r_{M_0 M}}$ 在 $\Omega \setminus \overline{B(M_0, \varepsilon)}$ 上调和知 $\iint_{\partial\Omega \cup \partial B(M_0, \varepsilon)} \frac{\partial}{\partial \vec{n}} \left(\frac{1}{4\pi r_{M_0 M}} \right) dS = 0$, 即

$$\iint_{\partial\Omega} \frac{\partial}{\partial \vec{n}} \left(\frac{1}{4\pi r_{M_0 M}} \right) dS = - \iint_{\partial B(M_0, \varepsilon)} \frac{\partial}{\partial \vec{n}} \left(\frac{1}{4\pi r_{M_0 M}} \right) dS.$$

由于 $\vec{n}$ 的方向为 M 指向 M_0 , 故

$$\frac{\partial}{\partial \vec{n}} \left(\frac{1}{4\pi r_{M_0 M}} \right) = - \frac{d}{dr_{M_0 M}} \left(\frac{1}{4\pi r_{M_0 M}} \right) = \frac{1}{4\pi r_{M_0 M}^2}.$$

故

$$\iint_{\partial B(M_0, \varepsilon)} \frac{\partial}{\partial \vec{n}} \left(\frac{1}{4\pi r_{M_0 M}} \right) ds = \frac{1}{4\pi \varepsilon^2} \iint_{\partial B(M_0, \varepsilon)} 1 dS = 1.$$

从而有 $\iint_{\partial\Omega} \frac{\partial}{\partial \vec{n}} \left(\frac{1}{4\pi r_{M_0 M}} \right) dS = -1$. □

4. 试用格林函数法导出调和方程第二边值问题解的表达式.

解. 考虑调和方程第二边值问题

$$\begin{cases} \Delta u = 0 & \text{在 } \Omega \text{ 上} \\ \frac{\partial u}{\partial \vec{n}} \Big|_{\partial\Omega} = f. \end{cases}$$

则有

$$u(M_0) = -\frac{1}{4\pi} \iint_{\partial\Omega} \left[u(M) \frac{\partial}{\partial \vec{n}} \left(\frac{1}{r_{M_0 M}} \right) - \frac{1}{r_{M_0 M}} \frac{\partial u(M)}{\partial \vec{n}} \right] dS. \quad (3.2)$$

引入函数 $g(M, M_0)$, 其满足

$$\begin{cases} \Delta g(\cdot, M_0) = 0 & \text{在 } \Omega \text{ 上}, \\ \frac{\partial g(\cdot, M_0)}{\partial \vec{n}} \Big|_{\partial\Omega} = \frac{\partial}{\partial \vec{n}} \left(\frac{1}{4\pi r_{M_0 M}} \right). \end{cases}$$

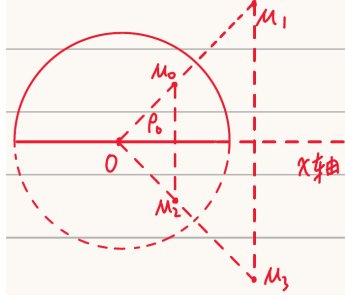
则有

$$\begin{aligned} 0 &= \iiint_{\Omega} [u(M) \Delta g(M, M_0) - g(M, M_0) \Delta u(M)] dx dy dz \\ &= \iint_{\partial\Omega} \left[u(M) \frac{\partial g(M, M_0)}{\partial \vec{n}} - g(M, M_0) \frac{\partial u(M)}{\partial \vec{n}} \right] dS. \end{aligned} \quad (3.3)$$

(3.2) 和 (3.3) 相加得

$$\begin{aligned} u(M_0) &= -\iint_{\partial\Omega} \left[u(M) \frac{\partial}{\partial \vec{n}} \left(\frac{1}{4\pi r_{M_0 M}} - g(M, M_0) \right) - \left(\frac{1}{4\pi r_{M_0 M}} - g(M, M_0) \right) \frac{\partial u(M)}{\partial \vec{n}} \right] dS \\ &= \iint_{\partial\Omega} G(M, M_0) f(M) dS, \end{aligned}$$

其中 $G(M, M_0) = \frac{1}{4\pi r_{M_0 M}} - g(M, M_0)$. □



解. 考虑如图半圆区域 $\Omega = B(O, R) \cap \mathbb{R}_+^3$. 任取 $M_0(x_0, y_0, z_0) \in \Omega$, M_1 是 M_0 关于球面 $\partial B(0, R)$ 的反演点, M_2 与 M_3 分别是 M_0 与 M_1 关于平面 $z = 0$ 的对称点, 则有

$$\overrightarrow{OM_1} = \frac{R^2}{\rho_0^2} \overrightarrow{OM_0}, \quad M_2 = (x_0, y_0, -z_0), \quad \overrightarrow{OM_3} = \frac{R^2}{\rho_0^2} \overrightarrow{OM_2}.$$

这也就是说, M_3 是 M_2 关于球面 $\partial B(0, R)$ 的反演点. 由上可知, 对 $\forall M \in \partial B(0, R)$, 有

$$r_{M_1 M} = \frac{R}{\rho_0} r_{M_0 M}, \quad r_{M_3 M} = \frac{R}{r_{OM_2}} r_{M_2 M} = \frac{R}{\rho_0} r_{M_2 M},$$

且对 $\forall M \in \partial \mathbb{R}_+^3$, 有 $r_{M_0 M} = r_{M_2 M}, r_{M_1 M} = r_{M_3 M}$. 令

$$G(M_1 M_0) = \frac{1}{2\pi} \left[\ln \frac{1}{r_{M_0 M}} - \ln \frac{1}{r_{M_2 M}} - \ln \left(\frac{R}{\rho_0} \frac{1}{r_{M_1 M}} \right) + \ln \left(\frac{R}{\rho_0} \frac{1}{r_{M_3 M}} \right) \right].$$

则由 M_1, M_2, M_3 均不在 Ω 上知 $\Delta G(M, M_0) = 0, \forall M \in \Omega$ 且 $M \neq M_0$. 对 $\forall M \in \partial \Omega \cap \partial B(0, R)$, 有

$$G(M, M_0) = \frac{1}{2\pi} \left[\left(\ln \frac{1}{r_{M_0 M}} - \ln \left(\frac{R}{\rho_0} \frac{1}{r_{M_1 M}} \right) \right) + \left(-\frac{1}{r_{M_2 M}} + \ln \left(\frac{R}{\rho_0} \frac{1}{r_{M_3 M}} \right) \right) \right] = 0;$$

对 $\forall M \in \partial \Omega \cap \partial \mathbb{R}_+^3$ 有

$$G(M, M_0) = \frac{1}{2\pi} \left[\left(\ln \frac{1}{r_{M_0 M}} - \ln \frac{1}{r_{M_2 M}} \right) + \left(-\ln \left(\frac{R}{\rho_0} \frac{1}{r_{M_1 M}} \right) + \ln \left(\frac{R}{\rho_0} \frac{1}{r_{M_3 M}} \right) \right) \right] = 0$$

故半圆区域 Ω 上狄利克雷问题的 Green 函数为

$$G(M, M_0) = \frac{1}{2\pi} \left[\ln \frac{1}{r_{M_0 M}} - \ln \frac{1}{r_{M_2 M}} - \ln \left(\frac{R}{\rho_0} \frac{1}{r_{M_1 M}} \right) + \ln \left(\frac{R}{\rho_0} \frac{1}{r_{M_3 M}} \right) \right].$$

□

9. 试求一函数 u , 使其在半径为 a 的圆的内部是调和的, 而且在圆周 C 上取下列的值:

(1) $u|_C = A \cos \theta,$

(2) $u|_C = A + B \sin \theta,$

其中 A, B 都是常数.

解. 圆盘 $B(O, a)$ 上泊松公式为

$$\begin{aligned} u(\rho_0, \theta_0) &= \frac{a^2 - \rho_0^2}{2\pi} \int_0^{2\pi} \frac{f(\theta)}{a^2 - 2a\rho_0 \cos(\theta - \theta_0) + \rho_0^2} d\theta \\ &= \frac{a^2 - \rho_0^2}{2\pi} \int_0^{2\pi} \frac{f(\theta + \theta_0)}{a^2 - 2a\rho_0 \cos \theta + \rho_0^2} d\theta, \end{aligned}$$

其中 $f(\theta) = u|_{\partial B(O,a)}$.

(1) 直接计算可得

$$\begin{aligned} u(\rho_0, \theta_0) &= \frac{A(a^2 - \rho_0^2)}{2\pi} \int_0^{2\pi} \frac{\cos(\theta + \theta_0)}{a^2 - 2a\rho_0 \cos \theta + \rho_0^2} d\theta \\ &= \frac{A(a^2 - \rho_0^2)}{2\pi} \left(\cos \theta_0 \int_0^{2\pi} \frac{\cos \theta}{a^2 - 2a\rho_0 \cos \theta + \rho_0^2} d\theta - \sin \theta_0 \int_0^{2\pi} \frac{\sin \theta}{a^2 - 2a\rho_0 \cos \theta + \rho_0^2} d\theta \right). \end{aligned}$$

一方面,

$$\begin{aligned} \int_0^{2\pi} \frac{\sin \theta}{a^2 - 2a\rho_0 \cos \theta + \rho_0^2} d\theta &= \frac{1}{2a\rho_0} \int_0^{2\pi} \frac{1}{a^2 - 2a\rho_0 \cos \theta + \rho_0^2} d(-2a\rho_0 \cos \theta) \\ &= \frac{1}{2a\rho_0} \ln(a^2 + \rho_0^2 - 2a\rho_0 \cos \theta) \Big|_0^{2\pi} = 0. \end{aligned}$$

另一方面, 由 Green 函数性质 5 知 $\frac{a^2 - \rho_0^2}{2\pi} \int_0^{2\pi} \frac{1}{a^2 - 2a\rho_0 \cos \theta + \rho_0^2} d\theta = 1$. 故

$$\begin{aligned} \int_0^{2\pi} \frac{\cos \theta}{a^2 - 2a\rho_0 \cos \theta + \rho_0^2} d\theta &= \frac{1}{2a\rho_0} \int_0^{2\pi} \left(-1 + \frac{a^2 + \rho_0^2}{a^2 - 2a\rho_0 \cos \theta + \rho_0^2} \right) d\theta \\ &= -\frac{\pi}{a\rho_0} + \frac{a^2 + \rho_0^2}{2a\rho_0} \frac{2\pi}{a^2 - \rho_0^2} \\ &= \frac{2\pi}{a^2 - \rho_0^2} \left(-\frac{a^2 - \rho_0^2}{2a\rho_0} + \frac{a^2 + \rho_0^2}{2a\rho_0} \right) \\ &= \frac{2\pi}{a^2 - \rho_0^2} \frac{\rho_0}{a} \end{aligned}$$

综上可得

$$u(\rho_0, \theta_0) = \frac{A(a^2 - \rho_0^2)}{2\pi} \frac{2\pi}{a^2 - \rho_0^2} \frac{\rho_0}{a} \cos \theta_0 = \frac{\rho_0}{a} A \cos \theta_0.$$

(2) 由 (1) 中计算结果可得

$$\begin{aligned} u(\rho_0, \theta_0) &= \frac{a^2 - \rho_0^2}{2\pi} A \int_0^{2\pi} \frac{1}{a^2 - 2a\rho_0 \cos \theta + \rho_0^2} d\theta + \frac{B(a^2 - \rho_0^2)}{2\pi} \int_0^{2\pi} \frac{\sin(\theta + \theta_0)}{a^2 - 2a\rho_0 \cos \theta + \rho_0^2} d\theta \\ &= A + \frac{B(a^2 - \rho_0^2)}{2\pi} \left(\cos \theta_0 \int_0^{2\pi} \frac{\sin \theta}{a^2 - 2a\rho_0 \cos \theta + \rho_0^2} d\theta + \sin \theta_0 \int_0^{2\pi} \frac{\cos \theta}{a^2 - 2a\rho_0 \cos \theta + \rho_0^2} d\theta \right) \\ &= A + \frac{\rho_0}{a} B \sin \theta_0. \end{aligned}$$

□

另解. 对 $\forall z = re^{i\theta} \in \mathbb{C}$ 且 $|z| < 1$, 有

$$\begin{aligned}
 \frac{1 - z\bar{z}}{1 - z - \bar{z} + z\bar{z}} &= \frac{1 - z + z(1 - \bar{z})}{(1 - z)(1 - \bar{z})} \\
 &= \frac{1}{1 - \bar{z}} + \frac{z}{1 - z} \\
 &= \sum_{n=0}^{+\infty} \bar{z}^n + \sum_{n=1}^{+\infty} z^n \\
 &= 1 + \sum_{n=1}^{+\infty} r^n (e^{-in\theta} + e^{in\theta}) \\
 &= 1 + 2 \sum_{n=1}^{+\infty} r^n \cos n\theta.
 \end{aligned}$$

取 $z = \frac{\rho_0}{a} e^{i\theta}$ ($\frac{\rho_0}{a} < 1$), 则有

$$\frac{a^2 - \rho_0^2}{a^2 - 2a\rho_0 \cos \theta + \rho_0^2} = \frac{1 - \left(\frac{\rho_0}{a}\right)^2}{1 - 2\frac{\rho_0}{a} \cos \theta + \left(\frac{\rho_0}{a}\right)^2} = \frac{1 - z\bar{z}}{1 - z - \bar{z} + z\bar{z}} = 1 + 2 \sum_{n=1}^{+\infty} \left(\frac{\rho_0}{a}\right)^n \cos n\theta.$$

另一方面, 注意到 $\int_0^{2\pi} \sin n\theta \cos m\theta d\theta = 0$, $\int_0^{2\pi} \cos n\theta \cos m\theta d\theta = 0$, $\forall n, m \in \mathbb{N}$ 且 $n \neq m$.

(1)

$$\begin{aligned}
 u(\rho_0, \theta_0) &= \frac{A}{2\pi} \int_0^{2\pi} \frac{a^2 - \rho_0^2}{a^2 - 2a\rho_0 \cos \theta + \rho_0^2} \cos(\theta + \theta_0) d\theta \\
 &= \frac{A}{2\pi} \int_0^{2\pi} (\cos \theta \cos \theta_0 - \sin \theta \sin \theta_0) \left[1 + 2 \sum_{n=1}^{+\infty} \left(\frac{\rho_0}{a}\right)^n \cos n\theta \right] d\theta \\
 &= \frac{A}{\pi} \frac{\rho_0}{a} \cos \theta_0 \int_0^{2\pi} \cos^2 \theta d\theta \\
 &= \frac{A}{\pi} \frac{\rho_0}{a} \cos \theta_0 \int_0^{2\pi} \frac{1 + \cos 2\theta}{2} d\theta \\
 &= \frac{\rho_0}{a} A \cos \theta_0.
 \end{aligned}$$

(2)

$$\begin{aligned}
 u(\rho_0, \theta_0) &= A + \frac{B}{2\pi} \int_0^{2\pi} (\sin \theta \cos \theta_0 + \cos \theta \sin \theta_0) \left[1 + 2 \sum_{n=1}^{+\infty} \left(\frac{\rho_0}{a}\right)^n \cos n\theta \right] d\theta \\
 &= A + \frac{B}{\pi} \frac{\rho_0}{a} \sin \theta_0 \int_0^{2\pi} \cos^2 \theta d\theta \\
 &= A + \frac{\rho_0}{a} B \sin \theta_0.
 \end{aligned}$$

□

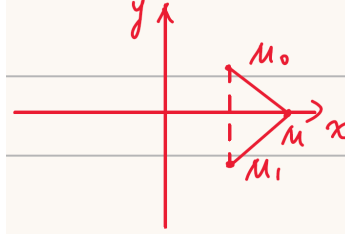
10. 试用静电像法导出二维调和方程在半平面上的狄利克雷问题:

$$\Delta u = u_{xx} + u_{yy} = 0, \quad y > 0,$$

$$u|_{y=0} = f(x)$$

的解.

解. 考虑半平面 $\mathbb{R}_+^2 = \{(x, y) \in \mathbb{R}^2 : y > 0\}$. 任取 $M_0(x_0, y_0) \in \mathbb{R}_+^2$, 取其关于 x 轴的对称点



$M_1(x_0, -y_0)$. 则对 $\forall M \in \partial\mathbb{R}_+^2$ 有 $r_{M_0M} = r_{M_1M}$. 令

$$\begin{aligned} G(M, M_0) &= \frac{1}{2\pi} \left(\ln \frac{1}{r_{M_0M}} - \ln \frac{1}{r_{M_1M}} \right) \\ &= \frac{1}{2\pi} \left(\ln \frac{1}{\sqrt{(x-x_0)^2 + (y-y_0)^2}} - \ln \frac{1}{\sqrt{(x-x_0)^2 + (y+y_0)^2}} \right). \end{aligned}$$

则 $\Delta G(M, M_0) = 0$, $\forall M \in \mathbb{R}_+^2$ 且 $M \neq M_0$. 对 $\forall M \in \partial\mathbb{R}_+^2$, 有 $G(M, M_0) = 0$. 故 $G(M, M_0)$ 是 $\mathbb{R}_+^2$ 上的 Green 函数. 对 $\forall M(x, y) \in \partial\mathbb{R}_+^2$, 有

$$\begin{aligned} \frac{\partial G(M, M_0)}{\partial \vec{n}} &= - \left. \frac{\partial G(M, M_0)}{\partial y} \right|_{y=0} \\ &= \frac{1}{2\pi} \left[\frac{1}{2} \frac{-2y_0}{(x-x_0)^2 + y_0^2} - \frac{1}{2} \frac{2y_0}{(x-x_0)^2 + y_0^2} \right] \\ &= -\frac{y_0}{\pi} \frac{1}{(x-x_0)^2 + y_0^2}. \end{aligned}$$

故

$$u(x_0, y_0) = - \int_{\partial\mathbb{R}_+^2} f(M) \frac{\partial G(M, M_0)}{\partial \vec{n}} ds = \frac{y_0}{\pi} \int_{-\infty}^{+\infty} \frac{f(x)}{(x-x_0)^2 + y_0^2} dx.$$

□

5 第 5 次作业

1. 试用强极值原理来证明极值原理.

证. 只需证明调和函数 u 在 Ω 上达不到最大值即可.

反证而设 u 在 $M_0 \in \Omega$ 上达到最大值 m , 即 $u(M_0) = m$. 则由 M_0 是 Ω 的内点知 $\nabla u(M_0) = 0$. 任意作球 $B \subset \Omega$, 使 $M_0 \in \partial B$. 则必有 $u|_{\bar{B}} \equiv m$. 否则, 若 u 在 $\bar{B}$ 上不恒为常数, 则由 Hopf 极值原理可知, 因为 u 在 ∂B 上的 M_0 取得其在 $\bar{B}$ 上的最大值, 成立 $\frac{\partial u}{\partial \vec{n}}(M_0) \neq 0$. 这与 $\nabla u(M_0) = 0$ 矛盾.

对 $\forall M_1 \in \Omega$, 由 Ω 连通过知, 存在有限长折线 γ 连接 M_0 与 M_1 . 于是存在有限个开球 $B_1, B_2, \dots, B_n$, 使得 $M_0 \in \partial B_1$, $M_1 \in B_n$, $\gamma \subset \bar{B}_1 \cup (\bigcup_{k=2}^n B_k)$ 且 $B_i \cap B_j \neq \emptyset$. 由上面分析可知 $u|_{\bar{B}_1} \equiv m$. 由 $B_1 \cap B_2 \neq \emptyset$ 可得 ∂B_2 上的最大值为 m , 从而再借用上述分析结果可得 $u|_{\bar{B}_2} \equiv m$. 依此类推, 得 $u|_{\bar{B}_n} \equiv m$. 特别地, 由于 $M_1 \in B_n$, 有 $u(M_1) = m$. 由 M_1 的任意性可得 $u|_{\Omega} \equiv m$. 这与 u 在 Ω 上不恒为常数矛盾. 因此 u 在 Ω 上无法达到最大值. $\square$

2. 考虑如下 Poisson 方程第三边值问题解的唯一性:

$$\begin{cases} -\Delta u = f & \text{在 } \Omega \text{ 上,} \\ \left(\frac{\partial u}{\partial \vec{n}} + \sigma u \right) \Big|_{\Gamma} = f, \end{cases} \quad (5.1)$$

其中 $\sigma > 0$, Ω 为满足内切球条件的有界区域.

(1) 利用极值原理及强极值原理证明 (5.1) 在 $C^2(\Omega) \cap C(\bar{\Omega})$ 上至多只有一个解.

(2) 利用能量法证明 (5.1) 在 $C^2(\Omega) \cap C^1(\bar{\Omega})$ 上至多只有一个解.

证. 设 u_1, u_2 是 (5.1) 的解. 令 $w = u_1 - u_2$, 则有

$$\begin{cases} -\Delta w = 0 & \text{在 } \Omega \text{ 上,} \\ \left(\frac{\partial w}{\partial \vec{n}} + \sigma w \right) \Big|_{\partial \Omega} = 0. \end{cases} \quad (5.2)$$

(1) 由于 w 是调和函数, 因此由极值原理可知 w 在边界 $\partial \Omega$ 上取到在 $\bar{\Omega}$ 上的最大最小值. 记最大值点与最小值点分别为 $M_1, M_2 \in \partial \Omega$. 于是有

$$\begin{cases} w(M_1) = -\frac{1}{\sigma} \frac{\partial w}{\partial \vec{n}}(M_1) \leq 0, \\ w(M_2) = -\frac{1}{\sigma} \frac{\partial w}{\partial \vec{n}}(M_2) \geq 0. \end{cases}$$

因此 $w(M_1) \leq 0 \leq w(M_2)$, 等号成立当且仅当 $\frac{\partial w}{\partial \vec{n}}(M_1) = \frac{\partial w}{\partial \vec{n}}(M_2) = 0$. 又 $u(M_1)$ 与 $u(M_2)$ 分别是 u 在 Ω 上的最大最小值, 因此 $w(M_1) \geq w(M_2)$. 从而 $w(M_1) = 0 = w(M_2)$, 即上述不等式取等号. 因此 $\frac{\partial w}{\partial \vec{n}}(M_1) = \frac{\partial w}{\partial \vec{n}}(M_2) = 0$. 于是由 Hopf 极值原理可知 $w|_{\bar{\Omega}} \equiv C$ (常数). 进而由第三类边界条件可得

$$\frac{\partial C}{\partial \vec{n}} + \sigma C = 0,$$

即 $C = 0$. 故 $w|_{\bar{\Omega}} \equiv 0$, 即在 Ω 上 $u_1 = u_2$. 唯一性得证.

(2) 定义能量

$$E(v) = \frac{1}{2} \iiint_{\Omega} |\nabla v|^2 dx dy dz + \frac{1}{2} \iint_{\partial\Omega} \sigma v^2 dS, \quad \forall v \in C^2(\Omega) \cap C'(\bar{\Omega}).$$

于是由 Green 公式以及 (5.2) 可得

$$\begin{aligned} E(w) &= \frac{1}{2} \iiint_{\Omega} |\nabla w|^2 dx dy dz + \frac{1}{2} \iint_{\partial\Omega} \sigma w^2 dS \\ &= -\frac{1}{2} \iiint_{\Omega} w \Delta w + \frac{1}{2} \iint_{\partial\Omega} \left(\frac{\partial w}{\partial \vec{n}} + \sigma w \right) w dS \\ &= 0 \end{aligned}$$

因此在 Ω 上恒有 $\nabla w = 0$, 在 $\partial\Omega$ 上恒有 $w = 0$. 由 $\nabla w = 0$ 知 $w|_{\Omega} \equiv C$ (常数). 结合 $w|_{\partial\Omega} = 0$ 可得 $w|_{\bar{\Omega}} = 0$. 因此在 $\bar{\Omega}$ 上 $u_1 = u_2$. 唯一性得证. $\square$

8. 设 Ω 为平面上的椭圆环 $\{(x, y) \mid 1 \leq x^2 + 2y^2 \leq 2\}$, $u(x, y) \in C^2(\bar{\Omega})$ 是如下的边值问题的解:

$$\begin{cases} \Delta u = 0, & (x, y) \in \Omega \\ u(x, y) = x + y, & x^2 + 2y^2 = 2 \\ \frac{\partial u(x, y)}{\partial \vec{n}} + (1 - x)u(x, y) = 0, & x^2 + 2y^2 = 1 \end{cases}$$

求 $\max_{\bar{\Omega}} |u(x, y)|$.

解. 记 $\Gamma_1 = \{(x, y) : x^2 + 2y^2 = 1\}$, $\Gamma_2 = \{(x, y) : x^2 + 2y^2 = 2\}$. 则由极值原理可知

$$\begin{cases} \max_{\bar{\Omega}} u = \max_{\partial\Omega} u = \max \left\{ \max_{\Gamma_1} u, \max_{\Gamma_2} u \right\}, \\ \min_{\bar{\Omega}} u = \min_{\partial\Omega} u = \min \left\{ \min_{\Gamma_1} u, \min_{\Gamma_2} u \right\}. \end{cases}$$

先计算 $\max_{\Gamma_2} u$ 与 $\min_{\Gamma_2} u$. 对 $\forall (x, y) \in \Gamma_2$, 由 Cauchy 不等式可得

$$|u(x, y)| = |x + y| = \left| x \cdot 1 + \sqrt{2}y \cdot \frac{1}{\sqrt{2}} \right| \leq \sqrt{(x^2 + 2y^2) \left(1 + \frac{1}{2} \right)} = \sqrt{3}.$$

等号成立当且仅当 $\frac{x}{1} = \frac{\sqrt{2}y}{\frac{1}{\sqrt{2}}}$, 即 $x = 2y$. 又 $x^2 + 2y^2 = 2$, 故等号成立当且仅当 $(x, y) =$

$\left(\frac{2}{3}\sqrt{3}, \frac{\sqrt{3}}{3} \right)$ 或 $\left(-\frac{2}{3}\sqrt{3}, -\frac{\sqrt{3}}{3} \right)$. 当 $(x, y) = \left(\frac{2}{3}\sqrt{3}, \frac{\sqrt{3}}{3} \right)$ 时, u 取得在 Γ_2 上的最大值, 即

$\max_{\Gamma_2} u = \sqrt{3}$. 当 $(x, y) = \left(-\frac{2}{3}\sqrt{3}, -\frac{\sqrt{3}}{3} \right)$ 时, u 取得在 Γ_2 上的最小值, 即 $\min_{\Gamma_2} u = -\sqrt{3}$.

下面说明 $\max_{\Gamma_1} u \leq \max_{\Gamma_2} u$ 且 $\min_{\Gamma_1} u \geq \min_{\Gamma_2} u$.

若 $\max_{\Gamma_1} u > \max_{\Gamma_2} u$, 即 u 在 Γ_1 上取得在 $\partial\Omega$ 上的最大值. 记最大值点为 $(x_1, y_1) \in \Gamma_1$. 于是有

$$(1 - x_1)u(x_1, y_1) = -\frac{\partial u}{\partial \vec{n}}(x_1, y_1).$$

显然, u 在 Ω 上不恒为常数, 因此由 Hopf 极值原理知 $\frac{\partial u}{\partial \vec{n}}(x_1, y_1) > 0$. 从而 $(1 - x_1)u(x_1, y_1) < 0$. 由 $x_1^2 + 2y_1^2 = 1$ 得 $|x_1| \leq 1$, 故 $1 - x_1 \geq 0$. 因此 $u(x_1, y_1) < 0 < \sqrt{3} = \max_{\Gamma_2} u$. 这与 $\max_{\Gamma_1} u > \max_{\Gamma_2} u$ 矛盾. 故 $\max_{\Gamma_1} u \leq \max_{\Gamma_2} u$.

若 $\min_{\Gamma_1} u < \min_{\Gamma_2} u$, 即 u 在 Γ_1 上取得在 $\partial\Omega$ 上的最小值. 记最小值点为 $(x_2, y_2) \in \Gamma_1$. 于是由 Hopf 极值原理可得

$$(1 - x_2)u(x_2, y_2) = -\frac{\partial u}{\partial \vec{n}}(x_2, y_2) > 0.$$

因此 $u(x_2, y_2) > 0 > -\sqrt{3} = \min_{\Gamma_2} u$. 这与 $\min_{\Gamma_1} u < \min_{\Gamma_2} u$ 矛盾. 故 $\min_{\Gamma_1} u \geq \min_{\Gamma_2} u$.

综上可得 $\max_{\bar{\Omega}} u = \max_{\Gamma_2} u = \sqrt{3}$, $\min_{\bar{\Omega}} u = \min_{\Gamma_2} u = -\sqrt{3}$. 故 $\max_{\bar{\Omega}} |u| = \sqrt{3}$. $\square$

6 第 6 次作业

1. 用分离变量法求下列定解问题的解:

$$\begin{cases} \frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2} & (t > 0, 0 < x < \pi), \\ u(0, t) = \frac{\partial u}{\partial x}(\pi, t) = 0 & (t > 0), \\ u(x, 0) = f(x) & (0 < x < \pi). \end{cases}$$

解. 令 $u(x, t) = X(x)T(t)$, 则有

$$\begin{cases} T'(t) + \lambda a^2 T(t) = 0, \\ X''(x) + \lambda X(x) = 0. \end{cases}$$

由边界条件知 $X(0)T(t) = X'(\pi)T(t) = 0$, 即 $X(0) = X'(\pi) = 0$.

(1) 当 $\lambda < 0$ 时, $X(x) = Ae^{\sqrt{-\lambda}x} + Be^{-\sqrt{-\lambda}x}$. 由 $X(0) = X'(\pi) = 0$ 得 $A = B = 0$, 即无非零解.

(2) 当 $\lambda = 0$ 时, $X(x) = A + Bx$. 由 $X(0) = X'(\pi) = 0$ 得 $A = B = 0$, 无非零解.

(3) 当 $\lambda > 0$ 时, $X(x) = A \cos \sqrt{\lambda}x + B \sin \sqrt{\lambda}x$. 由 $X(0) = X'(\pi) = 0$ 得

$$\begin{cases} A = 0, \\ B\sqrt{\lambda} \cos(\sqrt{\lambda}\pi) = 0. \end{cases}$$

为使 $B \neq 0$, 需 $\cos(\sqrt{\lambda}\pi) = 0$. 故 $\sqrt{\lambda}\pi = \frac{\pi}{2} + k\pi, k = 0, 1, 2, \dots$. 因此 $\lambda_k = \left(\frac{2k+1}{2}\right)^2, k = 0, 1, 2, \dots$ 相应地,

$$X_k = B_k \sin\left(\frac{2k+1}{2}x\right), k = 0, 1, 2, \dots$$

由 T 的方程知

$$T_k(t) = C_k e^{-\frac{a^2(2k+1)^2}{4}t}, k = 0, 1, 2, \dots$$

于是得到齐次方程齐次边界条件的一系列特解:

$$u_k(x, t) = A_k e^{-\frac{a^2(2k+1)^2}{4}t} \sin\left(\frac{2k+1}{2}x\right), k = 0, 1, 2, \dots$$

由叠加原理构造级数解:

$$u(x, t) = \sum_{k=0}^{+\infty} A_k e^{-\frac{a^2(2k+1)^2}{4}t} \sin\left(\frac{2k+1}{2}x\right).$$

由初始条件得

$$f(x) = u(x, 0) = \sum_{k=0}^{+\infty} A_k \sin\left(\frac{2k+1}{2}x\right).$$

由于 $\left\{\sin\left(\frac{2k+1}{2}x\right)\right\}_{k=0}^{+\infty}$ 构成 $[0, \pi]$ 上的正交系且

$$\int_0^\pi \sin^2\left(\frac{2k+1}{2}x\right) dx = \frac{1}{2} \int_0^\pi [1 - \cos(2k+1)x] dx = \frac{\pi}{2},$$

因此

$$A_k = \frac{2}{\pi} \int_0^\pi f(x) \sin \frac{2k+1}{2} x dx.$$

□

2. 用分离变量法求解热传导方程的初边值问题:

$$\begin{cases} \frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} & (t > 0, 0 < x < 1), \\ u(x, 0) = \begin{cases} x, & 0 < x \leq \frac{1}{2}, \\ 1-x, & \frac{1}{2} < x < 1, \end{cases} \\ u(0, t) = u(1, t) = 0 & (t > 0). \end{cases}$$

解. 由书本 P70 的 (2.11) 知题中齐次方程齐次 Dirichlet 边界条件的级数解为

$$u(x, t) = \sum_{k=1}^{+\infty} A_k e^{-a^2 k^2 \pi^2 t} \sin(k\pi x),$$

其中

$$\begin{aligned} A_k &= 2 \int_0^1 u(x, 0) \sin(k\pi x) dx \\ &= 2 \left[\int_0^{\frac{1}{2}} x \sin(k\pi x) dx + \int_{\frac{1}{2}}^1 (1-x) \sin(k\pi x) dx \right] \\ &= 2 \left[\int_0^{\frac{1}{2}} x \sin(k\pi x) dx + \int_{\frac{1}{2}}^0 x \sin(k\pi - k\pi x) d(1-x) \right] \\ &= 2 \left[\int_0^{\frac{1}{2}} x \sin(k\pi x) dx + (-1)^{k+1} \int_0^{\frac{1}{2}} x \sin(k\pi x) dx \right] \\ &= 2(1 + (-1)^{k+1}) \int_0^{\frac{1}{2}} x \sin(k\pi x) dx \\ &= \begin{cases} 4 \int_0^{\frac{1}{2}} x \sin(k\pi x) dx, & k \text{ 为奇数,} \\ 0, & k \text{ 为偶数.} \end{cases} \end{aligned}$$

直接计算可得

$$\begin{aligned} \int_0^{\frac{1}{2}} x \sin[(2k+1)\pi x] dx &= -\frac{1}{(2k+1)\pi} \int_0^{\frac{1}{2}} x d \cos[(2k+1)\pi x] \\ &= -\frac{1}{(2k+1)\pi} \left\{ x \cos[(2k+1)\pi x] \Big|_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} \cos[(2k+1)\pi x] dx \right\} \\ &= \frac{1}{(2k+1)^2 \pi^2} \sin[(2k+1)\pi x] \Big|_0^{\frac{1}{2}} \\ &= \frac{(-1)^k}{(2k+1)^2 \pi^2}. \end{aligned}$$

因此 $A_{2k} = 0$, $A_{2k+1} = (-1)^k \frac{4}{(2k+1)^2 \pi^2}$, $k = 0, 1, \dots$ 故原初边值问题的解为

$$u(x, t) = \sum_{k=0}^{+\infty} (-1)^k \frac{4}{(2k+1)^2 \pi^2} e^{-a^2(2k+1)^2 \pi^2 t} \sin[(2k+1)\pi x].$$

□

注 6.1. 上述积分也可通过以下方式来求解. 由

$$\begin{aligned} \int_0^{\frac{1}{2}} x \sin(k\pi x) dx &= -\frac{1}{k\pi} \left[x \cos(k\pi x) \Big|_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} \cos(k\pi x) dx \right] \\ &= -\frac{1}{k\pi} \cos \frac{k\pi}{2} + \frac{1}{(k\pi)^2} \sin \frac{k\pi}{2} \end{aligned}$$

以及

$$\begin{aligned} \int_{\frac{1}{2}}^1 (1-x) \sin(k\pi x) dx &= -\frac{1}{k\pi} \left[(1-x) \cos(k\pi x) \Big|_{\frac{1}{2}}^1 - \int_{\frac{1}{2}}^1 \cos(k\pi x) d(1-x) \right] \\ &= \frac{1}{k\pi} \cos \frac{k\pi}{2} - \frac{1}{k\pi} \int_{\frac{1}{2}}^1 \cos(k\pi x) dx \\ &= \frac{1}{k\pi} \cos \frac{k\pi}{2} - \frac{1}{(k\pi)^2} \sin(k\pi x) \Big|_{\frac{1}{2}}^1 \\ &= \frac{1}{k\pi} \cos \frac{k\pi}{2} + \frac{1}{(k\pi)^2} \sin \frac{k\pi}{2} \end{aligned}$$

得

$$\begin{aligned} A_k &= 2 \left[\int_0^{\frac{1}{2}} x \sin(k\pi x) dx + \int_{\frac{1}{2}}^1 (1-x) \sin(k\pi x) dx \right] \\ &= \frac{4}{(k\pi)^2} \sin \frac{k\pi}{2} \\ &= \begin{cases} (-1)^n \frac{4}{(2n+1)^2 \pi^2}, & k = 2n+1, \\ 0, & k = 2n. \end{cases} \end{aligned}$$

4. 在区域 $t > 0, 0 < x < l$ 中求解如下的定解问题:

$$\begin{cases} \frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2} - \beta(u - u_0), \\ u(0, t) = u(l, t) = u_0, \\ u(x, 0) = f(x), \end{cases}$$

其中 α, β, u_0 均为常数, $f(x)$ 为已知函数. (提示: 作未知函数代换 $u = u_0 + v(x, t)e^{-\beta t}$.)

解. 令 $u = u_0 + v(x, t)e^{-\beta t}$, 代入方程得

$$v_t e^{-\beta t} - \beta v e^{-\beta t} = a^2 v_{xx} e^{-\beta t} - \beta v e^{-\beta t},$$

即 $v_t = a^2 v_{xx}$. 代入边界条件得 $v(0, t)e^{-\beta t} = v(l, t)e^{-\beta t} = 0$, 故 $v(0, t) = v(l, t) = 0$. 代入初始条件得 $u_0 + v(x, 0) = f(x)$, 即 $v(x, 0) = f(x) - u_0$. 因此 v 是如下初边值问题的解:

$$\begin{cases} v_t = a^2 v_{xx}, & t > 0, 0 < x < l, \\ v(x, 0) = f(x) - u_0, & 0 < x < l, \\ v(0, t) = v(l, t) = 0, & t > 0. \end{cases}$$

从而

$$v(x, t) = \sum_{k=1}^{+\infty} A_k e^{-\frac{a^2 k^2 \pi^2}{l^2} t} \sin \frac{k\pi}{l} x,$$

其中

$$\begin{aligned} A_k &= \frac{2}{l} \int_0^l (f(x) - u_0) \sin \frac{k\pi}{l} x dx \\ &= \frac{2}{l} \int_0^l f(x) \sin \frac{k\pi}{l} x dx + \frac{2u_0}{l} \frac{l}{k\pi} \cos \frac{k\pi}{l} x \Big|_0^l \\ &= \frac{2}{l} \int_0^l f(x) \sin \frac{k\pi}{l} x dx + \frac{2u_0}{k\pi} [(-1)^k - 1]. \end{aligned}$$

故

$$u(x, t) = u_0 + v(x, t)e^{-\beta t} = u_0 + \sum_{k=1}^{+\infty} A_k e^{-\left(\beta + \frac{a^2 k^2 \pi^2}{l^2}\right)t} \sin \frac{k\pi}{l} x.$$

□

6. 半径为 a 的半圆形平板, 其表面绝热, 在板的圆周边界上保持常温 u_0 , 而在直径边界上保持常温 u_1 , 求圆板稳恒状态 (即与时间 t 无关的状态) 的温度分布.

解. 记 $\Omega = \{(x, y) \in \mathbb{R}^2 : \sqrt{x^2 + y^2} < a \text{ 且 } y > 0\}$, $\Gamma_1 = \{(x, y) \in \mathbb{R}^2 : \sqrt{x^2 + y^2} = a \text{ 且 } y > 0\}$, $\Gamma_2 = \{(x, 0) \in \mathbb{R}^2 : -a \leq x \leq a\}$.

稳态温度分布 $u(x, y)$ 满足如下泊松方程的 Dirichlet 边值问题:

$$\begin{cases} \Delta u = 0 & \text{在 } \Omega \text{ 上} \\ u|_{\Gamma_1} = u_0, \\ u|_{\Gamma_2} = u_1. \end{cases} \quad (6.1)$$

作极坐标变换: $(x, y) \rightarrow (r, \theta)$. 由 P106 的习题 3 知

$$\Delta u = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2}.$$

将 u 看成 r, θ 的函数, 记为 $u(r, \theta)$. 于是区域 Ω 转化为了 $(0, a) \times (0, \pi)$, Γ_1 转化为了 $\{a\} \times (0, \pi)$, Γ_2 转化为了 $[0, a] \times \{0, \pi\}$. 从而 $u|_{\Gamma_1} = u_0$ 转换为了 $u(a, \theta) = u_0, \forall \theta \in (0, \pi)$; $u|_{\Gamma_2} = u_1$ 转化为了 $u(r, 0) = u(r, \pi) = u_1$. 因此问题 (6.1) 转化为了

$$\begin{cases} u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} = 0, & 0 < r < a, 0 < \theta < \pi \\ u(a, \theta) = u_0, & 0 < \theta < \pi, \\ u(r, 0) = u(r, \pi) = u_1, & 0 \leq r \leq a. \end{cases}$$

令 $u = v + u_1$, 则方程转化为

$$\begin{cases} v_{rr} + \frac{1}{r}v_r + \frac{1}{r^2}v_{\theta\theta} = 0, & 0 < r < a, 0 < \theta < \pi, \\ v(a, \theta) = u_0 - u_1, & 0 < \theta < \pi, \\ v(r, 0) = v(r, \pi) = 0, & 0 \leq r \leq a. \end{cases}$$

令 $v(r, \theta) = R(r)\Theta(\theta)$. 代入方程得

$$\Theta(\theta)R''(r) + \frac{1}{r}\Theta(\theta)R'(r) + \frac{1}{r^2}\Theta''(\theta)R(r) = 0,$$

从而有

$$\frac{R''(r) + \frac{1}{r}R'(r)}{\frac{1}{r^2}R(r)} = -\frac{\Theta''(\theta)}{\Theta(\theta)} = \lambda,$$

即

$$\begin{cases} R''(r) + \frac{1}{r}R'(r) - \frac{\lambda}{r^2}R(r) = 0, \\ \Theta''(\theta) + \lambda\Theta(\theta) = 0. \end{cases}$$

由 $V(r, 0) = V(r, \pi) = 0$ 知 $\Theta(0) = \Theta(\pi) = 0$. 由 Θ 的方程知

$$\Theta(\theta) = \begin{cases} Ae^{\sqrt{-\lambda}\theta} + Be^{-\sqrt{-\lambda}\theta}, & \lambda < 0, \\ A + B\theta, & \lambda = 0, \\ A \cos \sqrt{\lambda}\theta + B \sin \sqrt{\lambda}\theta, & \lambda > 0. \end{cases}$$

结合边界条件知 $\lambda < 0$ 与 $\lambda = 0$ 均无非零解. 考虑 $\lambda > 0$, 得

$$\begin{cases} A = 0, \\ B \sin(\sqrt{\lambda}\pi) = 0. \end{cases}$$

故 $\sqrt{\lambda}\pi = k\pi, k = 1, 2, \dots$ 于是得到 $\lambda_k = k^2, k = 1, 2, \dots$, 相应地,

$$\Theta_k = B_k \sin(k\theta), k = 1, 2, \dots$$

由 R_k 有界设 $R_k(r) = \sum_{n=0}^{+\infty} a_n r^n$. 由 $R_k''(r) + \frac{1}{r}R_k'(r) - \frac{\lambda_k}{r^2}R_k(r) = 0$ 得

$$\sum_{n=-\infty}^{+\infty} a_n [n(n-1) + n - k^2] r^{n-2} = 0,$$

即

$$\sum_{n=-\infty}^{+\infty} a_n (n^2 - k^2) r^{n-2} = 0.$$

于是只有 $n = k$ 时 $a_n \neq 0$, 从而 $R_k(r) = C_k r^k$. 故得到一列特解 $v_k(r, \theta) = A_k r^k \sin(k\theta)$. 由叠加原理构造级数解:

$$v(r, \theta) = \sum_{k=1}^{+\infty} A_k r^k \sin(k\theta).$$

由 $v(a, \theta) = u_0 - u_1$ 得 $u_0 - u_1 = \sum_{k=1}^{+\infty} A_k a^k \sin(k\theta)$. 于是有

$$A_k a^k \int_0^\pi \sin^2(k\theta) d\theta = \int_0^\pi (u_0 - u_1) \sin(k\theta) d\theta = - \left. \frac{u_0 - u_1}{k} \cos(k\theta) \right|_0^\pi = \frac{u_0 - u_1}{k} [1 - (-1)^k].$$

由于 $\int_0^\pi \sin^2(k\theta) d\theta = \frac{\pi}{2}$, 故 $A_k = \frac{2}{\pi a^k} \frac{u_0 - u_1}{k} [1 - (-1)^k]$. 因此

$$v(r, \theta) = \frac{4(u_0 - u_1)}{\pi} \sum_{k=0}^{+\infty} \frac{1}{2k+1} \left(\frac{r}{a}\right)^{2k+1} \sin[(2k+1)\theta].$$

于是

$$u = v + u_1 = u_1 + \frac{4(u_0 - u_1)}{\pi} \sum_{k=0}^{+\infty} \frac{1}{2k+1} \left(\frac{r}{a}\right)^{2k+1} \sin[(2k+1)\theta].$$

□

7. 设 $u(x, t)$ 是 $(0, \frac{\pi}{2}) \times (0, \infty)$ 中初边值问题

$$\begin{cases} u_t = u_{xx}, \\ u(0, t) = 1, \quad u\left(\frac{\pi}{2}, t\right) = 4, \\ u(x, 0) = \cos^4 x + 4 \sin^5 x \end{cases}$$

的解, 求 $u(x, t)$.

解. 令 $u(x, t) = v(x, t) + \frac{6}{\pi}x + 1$, 则有

$$\begin{cases} v_t = v_{xx}, & t > 0, 0 < x < \frac{\pi}{2}, \\ v(0, t) = v\left(\frac{\pi}{2}, t\right) = 0, & t > 0, \\ v(x, 0) = \cos^4 x + 4 \sin^5 x - \frac{6}{\pi}x - 1, & 0 < x < \frac{\pi}{2}. \end{cases}$$

于是该初边值问题的级数解为

$$v(x, t) = \sum_{k=1}^{+\infty} A_k e^{-4k^2 a^2 t} \sin(2kx),$$

其中 $A_k = \frac{4}{\pi} \int_0^{\frac{\pi}{2}} \left(\cos^4 x + 4 \sin^5 x - \frac{6}{\pi}x - 1 \right) \sin(2kx) dx$, $k \geq 1$. 下面来一项项计算积分值.

先给出 $\cos^4 x$ 与 $\sin^5 x$ 的新表达式. 由 $\cos x = \frac{1}{2}(e^{ix} + e^{-ix})$ 得

$$\begin{aligned} \cos^4 x &= \frac{1}{16} \sum_{k=0}^4 \binom{4}{k} e^{i(4-k)x} e^{-ikx} \\ &= \frac{1}{16} \sum_{k=0}^4 \binom{4}{k} e^{i(4-2k)x} \\ &= \frac{1}{16} [e^{i4x} + e^{-i4x} + 4(e^{i2x} + e^{-i2x}) + 6] \\ &= \frac{1}{8} (\cos 4x + 4 \cos 2x + 3). \end{aligned}$$

由 $\sin x = \frac{1}{2i}(e^{ix} - e^{-ix})$ 得

$$\begin{aligned}\sin^5 x &= \frac{1}{32i} \sum_{k=0}^5 (-1)^k \binom{5}{k} e^{i(5-2k)x} \\ &= \frac{1}{32i} [e^{i5x} - e^{-i5x} + 5(-e^{i3x} + e^{-i3x}) + 10(e^{ix} - e^{-ix})] \\ &= \frac{1}{16} (\sin 5x - 5 \sin 3x + 10 \sin x).\end{aligned}$$

下面来计算积分.

对 $\forall n, k \in \mathbb{N}, k \geq 1$ 且 $n \neq k$, 有

$$\begin{aligned}& \int_0^{\frac{\pi}{2}} \cos(2nx) \sin(2kx) dx \\ &= \frac{1}{2} \int_0^{\frac{\pi}{2}} [\sin(2nx + 2kx) - \sin(2nx - 2kx)] dx \\ &= \frac{1}{2} \left[-\frac{1}{2(n+k)} \cos(2nx + 2kx) \Big|_0^{\frac{\pi}{2}} + \frac{1}{2(n-k)} \cos(2nx - 2kx) \Big|_0^{\frac{\pi}{2}} \right] \\ &= \frac{1}{4(n+k)} [1 - (-1)^{n+k}] + \frac{1}{4(n-k)} [(-1)^{n-k} - 1] \\ &= [1 - (-1)^{n+k}] \frac{k}{2(k^2 - n^2)},\end{aligned}$$

且由上式推导可知当 $n = k$ 时, 有

$$\int_0^{\frac{\pi}{2}} \cos(2kx) \sin(2kx) dx = \frac{1}{8k} [1 - (-1)^{2k}] = 0.$$

于是对 $\forall k \geq 3$, 有

$$\begin{aligned}& \int_0^{\frac{\pi}{2}} \cos^4 x \sin(2kx) dx \\ &= \frac{1}{8} \int_0^{\frac{\pi}{2}} (\cos 4x + 4 \cos 2x + 3) \sin(2kx) dx \\ &= \frac{1}{8} \left\{ [1 - (-1)^{2+k}] \frac{k}{2(k^2 - 4)} + [1 - (-1)^{1+k}] \frac{2k}{k^2 - 1} + [1 - (-1)^k] \frac{3k}{2k^2} \right\} \\ &= \frac{k}{8} \left\{ [1 - (-1)^k] \frac{1}{2(k^2 - 4)} + [1 + (-1)^k] \frac{2}{k^2 - 1} + [1 - (-1)^k] \frac{3}{2k^2} \right\}.\end{aligned}$$

由上面计算知当 $k = 1$ 时, 有

$$\int_0^{\frac{\pi}{2}} \cos^4 x \sin(2x) dx = \frac{1}{8} \left(-\frac{1}{3} + \frac{3}{1} \right) = \frac{1}{3}.$$

当 $k = 2$ 时, 有

$$\int_0^{\frac{\pi}{2}} \cos^4 x \sin(4x) dx = \frac{2}{8} \times \frac{4}{3} = \frac{1}{3}.$$

对 $\forall n, k \in \mathbb{N}$ 且 $k \geq 1$, 有

$$\begin{aligned}
& \int_0^{\frac{\pi}{2}} \sin[(2n+1)x] \sin(2kx) dx \\
&= \frac{1}{2} \int_0^{\frac{\pi}{2}} [\cos((2n+1-2k)x) - \cos((2n+1+2k)x)] dx \\
&= \frac{1}{2} \left[\frac{1}{2(n-k)+1} \sin((2n+1-2k)x) \Big|_0^{\frac{\pi}{2}} - \frac{1}{2(n+k)+1} \sin((2n+1+2k)x) \Big|_0^{\frac{\pi}{2}} \right] \\
&= \frac{(-1)^{n-k}}{4(n-k)+2} - \frac{(-1)^{n+k}}{4(n+k)+2} \\
&= (-1)^{n+k+1} \frac{2k}{4k^2 - (2n+1)^2}.
\end{aligned}$$

于是

$$\begin{aligned}
& \int_0^{\frac{3}{2}} \sin^5 x \sin(2kx) dx \\
&= \frac{1}{16} \int_0^{\frac{\pi}{2}} (\sin 5x - 5 \sin 3x + 10 \sin x) \sin(2kx) dx \\
&= \frac{1}{16} \left[(-1)^{2+k+1} \frac{2k}{4k^2-25} - (-1)^{1+k+1} \frac{10k}{4k^2-9} + (-1)^{k+1} \frac{20k}{4k^2-1} \right] \\
&= \frac{(-1)^{k+1}k}{8} \left(\frac{1}{4k^2-25} + \frac{5}{4k^2-9} + \frac{10}{4k^2-1} \right).
\end{aligned}$$

当 $k=1$ 时,

$$\int_0^{\frac{\pi}{2}} \sin^5 x \sin(2x) dx = \frac{1}{8} \left(-\frac{1}{21} - \frac{5}{5} + \frac{10}{3} \right) = \frac{2}{7}.$$

当 $k=2$ 时,

$$\int_0^{\frac{\pi}{2}} \sin^5 x \sin(4x) dx = -\frac{1}{4} \left(-\frac{1}{9} + \frac{5}{7} + \frac{10}{15} \right) = -\frac{20}{63}.$$

由

$$\int_0^{\frac{\pi}{2}} \sin(2kx) dx = -\frac{1}{2k} \cos(2kx) \Big|_0^{\frac{\pi}{2}} = \frac{1}{2k} [1 - (-1)^k]$$

以及

$$\begin{aligned}
\int_0^{\frac{\pi}{2}} x \sin(2kx) dx &= -\frac{1}{2k} \int_0^{\frac{\pi}{2}} x d \cos(2kx) \\
&= -\frac{1}{2k} \left[x \cos(2kx) \Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \cos(2kx) dx \right] \\
&= \frac{(-1)^{k+1}\pi}{4k} + \frac{1}{(2k)^2} \sin(2kx) \Big|_0^{\frac{\pi}{2}} \\
&= \frac{(-1)^{k+1}\pi}{4k}
\end{aligned}$$

得

$$\int_0^{\frac{\pi}{2}} \left(\frac{6}{\pi}x + 1 \right) \sin(2kx) dx = (-1)^{k+1} \frac{3}{2k} + \frac{1}{2k} [1 - (-1)^k].$$

当 $k = 1$ 时,

$$\int_0^{\frac{\pi}{2}} \left(\frac{6}{\pi}x + 1 \right) \sin 2x dx = \frac{3}{2} + \frac{2}{2} = \frac{5}{2}.$$

当 $k = 2$ 时,

$$\int_0^{\frac{\pi}{2}} \left(\frac{6}{\pi}x + 1 \right) \sin 4x dx = -\frac{3}{4}.$$

综上, 可得

$$A_1 = \frac{4}{\pi} \int_0^{\frac{\pi}{2}} \left(\cos^4 x + 4 \sin^5 x - \frac{6}{\pi}x - 1 \right) \sin 2x dx = \frac{4}{\pi} \left(\frac{1}{3} + \frac{2}{7} \times 4 - \frac{5}{2} \right) = -\frac{86}{21\pi},$$

$$A_2 = \frac{4}{\pi} \int_0^{\frac{\pi}{2}} \left(\cos^4 x + 4 \sin^5 x - \frac{6}{\pi}x - 1 \right) \sin 4x dx = \frac{4}{\pi} \left(\frac{1}{3} - \frac{20}{63} \times 4 + \frac{3}{4} \right) = -\frac{47}{63\pi}.$$

当 $k \geq 3$ 时,

$$\begin{aligned} A_k &= \frac{4}{2} \int_0^{\frac{\pi}{2}} \left(\cos^4 x + 4 \sin^5 x - \frac{6}{2}x - 1 \right) \sin(2kx) dx \\ &= \frac{k}{2\pi} \left\{ \frac{1 - (-1)^k}{2(k^2 - 4)} + \frac{2[1 + (-1)^k]}{k^2 - 1} + \frac{3[1 - (-1)^k]}{2k^2} \right\} \\ &\quad + (-1)^{k+1} \frac{2k}{\pi} \left(\frac{1}{4k^2 - 25} + \frac{5}{4k^2 - 9} + \frac{10}{4k^2 - 1} \right) - (-1)^{k+1} \frac{6}{k\pi} - \frac{2[1 - (-1)^k]}{k\pi}. \end{aligned}$$

□

7 第 7 次作业

1. 求下述函数的傅里叶变换:

(1) $e^{-\eta x^2} \quad (\eta > 0);$

(2) $e^{-a|x|} \quad (a > 0);$

(3) $\frac{x}{a^2 + x^2}, \frac{1}{a^2 + x^2}$

证. (1) 记 $f(x) = e^{-\eta x^2}$, 则

$$\begin{aligned} F[f](\lambda) &= \int_{-\infty}^{+\infty} e^{-\eta x^2} e^{-i\lambda x} dx \\ &= \int_{-\infty}^{+\infty} e^{-\eta(x^2 + i\frac{\lambda x}{\eta})} dx \\ &= e^{-\frac{\lambda^2}{4\eta}} \int_{-\infty}^{+\infty} e^{-\eta(x + i\frac{\lambda}{2\eta})^2} dx \\ &= e^{-\frac{\lambda^2}{4\eta}} \int_{-\infty}^{+\infty} e^{-\eta x^2} dx \\ &= \frac{1}{\sqrt{\eta}} e^{-\frac{\lambda^2}{4\eta}} \int_{-\infty}^{+\infty} e^{-y^2} dy \quad \left(\text{令 } x = \frac{1}{\sqrt{\eta}} y \right) \\ &= \frac{\sqrt{\pi}}{\sqrt{\eta}} e^{-\frac{\lambda^2}{4\eta}} \end{aligned}$$

(2) 记 $f(x) = e^{-a|x|}$, 则

$$\begin{aligned} F[f](\lambda) &= \int_{-\infty}^{+\infty} e^{-a|x|} e^{-i\lambda x} dx \\ &= \int_0^{+\infty} e^{-(a+i\lambda)x} dx + \int_{-\infty}^0 e^{(a-i\lambda)x} dx \\ &= -\frac{1}{a+i\lambda} e^{-(a+i\lambda)x} \Big|_0^{+\infty} + \frac{1}{a-i\lambda} e^{(a-i\lambda)x} \Big|_{-\infty}^0 \\ &= \frac{1}{a+i\lambda} + \frac{1}{a-i\lambda} \\ &= \frac{2a}{(a+i\lambda)(a-i\lambda)} \\ &= \frac{2a}{a^2 + \lambda^2}. \end{aligned}$$

(3) 记 $f(x) = \frac{1}{a^2+x^2}$, 则 $\frac{x}{a^2+x^2} = xf(x)$. 由 (2) 知

$$\begin{aligned} F^{-1} [e^{-a|\lambda|}] (x) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{-a|\lambda|} e^{i\lambda x} d\lambda \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{-a|\lambda|} e^{-i(-x)\lambda} d\lambda \\ &= \frac{1}{2\pi} F [e^{-a|\lambda|}] (-x) \\ &= \frac{1}{2\pi} \frac{2a}{a^2+x^2} \\ &= \frac{a}{\pi} \frac{1}{a^2+x^2} \end{aligned}$$

故 $F[f](\lambda) = \frac{\pi}{a} e^{-a|\lambda|}$. 从而有

$$F \left[\frac{x}{a^2+x^2} \right] (\lambda) = \frac{1}{-i} F[-ixf] = i \frac{d}{d\lambda} \hat{f}(\lambda) = -i\pi \operatorname{sign} \lambda e^{-a|\lambda|},$$

其中

$$\operatorname{sign} \lambda = \begin{cases} 1, & \lambda \geq 0, \\ -1, & \lambda < 0. \end{cases}$$

□

3. 用傅里叶变换求解三维热传导方程的柯西问题

$$\begin{cases} \frac{\partial u}{\partial t} = a^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right), \\ u|_{t=0} = \varphi(x, y, z). \end{cases}$$

解. 关于 x, y, z 作三维 Fourier 变换得

$$\begin{cases} \hat{u}_t = -a^2 (\lambda_1^2 + \lambda_2^2 + \lambda_3^2) \hat{u}. \\ \hat{u}(\lambda, 0) = \hat{\varphi}(\lambda). \end{cases}$$

故 $\hat{u}(\lambda, t) = \hat{\varphi}(\lambda) e^{-a^2(\lambda_1^2+\lambda_2^2+\lambda_3^2)t}$. 由习题 1(1) 知

$$\begin{aligned} & F^{-1} \left[e^{-a^2(\lambda_1^2+\lambda_2^2+\lambda_3^2)t} \right] (x, y, z) \\ &= \frac{1}{(2\pi)^3} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} e^{-a^2(\lambda_1^2+\lambda_2^2+\lambda_3^2)t} e^{i\lambda_1 x + i\lambda_2 y + i\lambda_3 z} d\lambda_1 d\lambda_2 d\lambda_3 \\ &= \frac{1}{(2\pi)^3} \int_{-\infty}^{+\infty} e^{-a^2\lambda_1^2 t} e^{i\lambda_1 x} d\lambda_1 \int_{-\infty}^{+\infty} e^{-a^2\lambda_2^2 t} e^{i\lambda_2 y} d\lambda_2 \int_{-\infty}^{+\infty} e^{-a^2\lambda_3^2 t} e^{i\lambda_3 z} d\lambda_3 \\ &= \frac{1}{(2\pi)^3} F \left[e^{-a^2 t \lambda_1^2} \right] (-x) F \left[e^{-a^2 t \lambda_2^2} \right] (-y) F \left[e^{-a^2 t \lambda_3^2} \right] (-z) \\ &= \frac{1}{(2\pi)^3} \left(\frac{\sqrt{\pi}}{a\sqrt{t}} \right)^3 e^{-\frac{x^2+y^2+z^2}{4a^2 t}} \\ &= \frac{1}{(4a^2 \pi t)^{\frac{3}{2}}} e^{-\frac{x^2+y^2+z^2}{4a^2 t}} \end{aligned}$$

故

$$\begin{aligned}
u(x, t) &= F^{-1}[\hat{u}](x, t) \\
&= \left(\varphi * F^{-1} \left[e^{-a^2(x_1^2 + \lambda_2^2 + \lambda_3^2)t} \right] \right)(x, t) \\
&= \frac{1}{(4a^2\pi t)^{\frac{3}{2}}} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \varphi(\xi_1, \xi_2, \xi_3) e^{-\frac{(x-\xi_1)^2 + (y-\xi_2)^2 + (z-\xi_3)^2}{4a^2t}} d\xi_1 d\xi_2 d\xi_3.
\end{aligned}$$

□

5. 求解热传导方程 (3.17) 的柯西问题, 已知 $u|_{t=0} = \sin x$.

解. 由 Poisson 公式以及习题 1(1) 的结论可得

$$\begin{aligned}
u(x, t) &= \frac{1}{2a\sqrt{\pi t}} \int_{-\infty}^{+\infty} \sin \xi e^{-\frac{(x-\xi)^2}{4a^2t}} d\xi \\
&= \frac{1}{2a\sqrt{\pi t}} \int_{-\infty}^{+\infty} \sin(x + \eta) e^{-\frac{\eta^2}{4a^2t}} d\eta \quad (\text{令 } \xi = x + \eta) \\
&= \frac{1}{2a\sqrt{\pi t}} \left(\sin x \int_{-\infty}^{+\infty} \cos \eta e^{-\frac{\eta^2}{4a^2t}} d\eta + \cos x \int_{-\infty}^{+\infty} \sin \eta e^{-\frac{\eta^2}{4a^2t}} d\eta \right) \\
&= \frac{\sin x}{4a\sqrt{\pi t}} \left(\int_{-\infty}^{+\infty} e^{-\frac{\eta^2}{4a^2t}} e^{i\eta} d\eta + \int_{-\infty}^{+\infty} e^{-\frac{\eta^2}{4a^2t}} e^{-i\eta} d\eta \right) \\
&= \frac{\sin x}{4a\sqrt{\pi t}} \left[F \left[e^{-\frac{\eta^2}{4a^2t}} \right] (-1) + F \left[e^{-\frac{\eta^2}{4a^2t}} \right] (1) \right] \\
&= \frac{\sin x}{4a\sqrt{\pi t}} 2a\sqrt{t}\sqrt{\pi} \times 2e^{-a^2t} \\
&= e^{-a^2t} \sin x.
\end{aligned}$$

□

7. 证明: 如果 $u_1(x, t), u_2(y, t)$ 分别是下述两个定解问题的解:

$$\begin{cases} \frac{\partial u_1}{\partial t} = a^2 \frac{\partial^2 u_1}{\partial x^2}, & \begin{cases} \frac{\partial u_2}{\partial t} = a^2 \frac{\partial^2 u_2}{\partial y^2}, \\ u_2|_{t=0} = \varphi_2(y). \end{cases} \\ u_1|_{t=0} = \varphi_1(x); \end{cases}$$

则 $u(x, y, t) = u_1(x, t)u_2(y, t)$ 是定解问题

$$\begin{cases} \frac{\partial u}{\partial t} = a^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right), \\ u|_{t=0} = \varphi_1(x)\varphi_2(y) \end{cases}$$

的解.

证. 直接计算可得 $\frac{\partial^2 u}{\partial x^2}(x, y, t) = \frac{\partial^2 u_1}{\partial x^2}(x, t)u_2(y, t)$, $\frac{\partial^2 u}{\partial y^2}(x, y, t) = u_1(x, t)\frac{\partial^2 u_2}{\partial y^2}(y, t)$, 以及

$$\frac{\partial u}{\partial t}(x, y, t) = \frac{\partial u_1}{\partial t}(x, t)u_2(y, t) + u_1(x, t)\frac{\partial u_2}{\partial t}(y, t) = a^2 \frac{\partial^2 u_1}{\partial x^2} u_2(y, t) + a^2 u_1(x, t) \frac{\partial^2 u_2}{\partial y^2}(y, t).$$

故

$$\frac{\partial u}{\partial t} - a^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) = 0.$$

又 $u(x, y, 0) = u_1(x, 0)u_2(y, 0) = \varphi_1(x)\varphi_2(y)$, 因此 $u(x, y, t)$ 是题设中定解问题的解.

□

附加题 1. 利用分离变量法和齐次化原理求解下列初边值问题:

$$\begin{cases} u_t - a^2 u_{xx} = f(x, t) & t > 0, 0 < x < l, \\ t = 0 : u = \varphi(x) & 0 < x < l, \\ x = 0 \text{ 及 } x = l : u = 0 & t > 0. \end{cases}$$

解. 将原初边值问题分为如下两个初边值问题

$$\begin{cases} u_t - a^2 u_{xx} = 0 & t > 0, 0 < x < l, \\ u(x, 0) = \varphi(x) & 0 < x < l, \\ u(0, t) = u(l, t) = 0 & t > 0 \end{cases} \quad (7.1)$$

和

$$\begin{cases} u_t - a^2 u_{xx} = f(x, t) & t > 0, 0 < x < l \\ u(x, 0) = 0 & 0 < x < l, \\ u(0, t) = u(l, t) = 0 & t > 0. \end{cases} \quad (7.2)$$

记 (7.1) 的解为 u_1 , (7.2) 的解为 u_2 , 则 $u = u_1 + u_2$ 是原初边值问题的解.

由分离变量法可得 (7.1) 的解为

$$u_1(x, t) = \sum_{k=1}^{+\infty} A_k e^{-\frac{a^2 k^2 x^2}{l} t} \sin \frac{kx}{l} x,$$

其中

$$A_k = \frac{2}{l} \int_0^l \varphi(\xi) \sin \frac{kx}{l} \xi d\xi.$$

下面求解 u_2 . 由齐次化原理知 $u_2(x, t) = \int_0^t w(x, t; \tau) d\tau$, 其中 $w(x, t; \tau)$ 满足

$$\begin{cases} w_t - a^2 w_{xx} = 0 & t > \tau, 0 < x < l, \\ w(x, \tau; \tau) = f(x, \tau) & 0 < x < l, \\ w(0, t; \tau) = w(l, t; \tau) = 0 & t > \tau. \end{cases}$$

令 $t' = t - \tau$, 有

$$\begin{cases} w_{t'} - a^2 w_{xx} = 0 & t' > 0, 0 < x < l, \\ t' = 0 : w = f(x, \tau) & 0 < x < l, \\ x = 0 \text{ 及 } x = l : w = 0 & t' > 0. \end{cases}$$

于是由 (1) 的解可知

$$w(x, t; \tau) = \sum_{k=1}^{+\infty} D_k(\tau) e^{-\frac{a^2 k^2 x^2}{l^2} t'} \sin \frac{k\pi}{l} x = \sum_{k=1}^{+\infty} D_k(\tau) e^{-\frac{a^2 k^2}{l} (t-\tau)} \sin \frac{k\pi}{l} x,$$

其中

$$D_k(\tau) = \frac{2}{l} \int_0^l f(\xi, \tau) \sin \frac{k\pi}{l} \xi d\xi.$$

因此有

$$u_2(x, t) = \sum_{k=1}^{+\infty} \int_0^t D_k(\tau) e^{-\frac{a^2 k^2 \pi^2}{l^2}(t-\tau)} d\tau \sin \frac{k\pi}{l} x.$$

故

$$u(x, t) = \sum_{k=1}^{+\infty} A_k e^{-\frac{a^2 k^2 \pi^2}{l^2} t} \sin \frac{k\pi}{l} x + \sum_{k=1}^{+\infty} \int_0^t D_k(\tau) e^{-\frac{a^2 k^2 \pi^2}{l^2}(t-\tau)} d\tau \sin \frac{k\pi}{l} x,$$

其中

$$A_k = \frac{2}{l} \int_0^l \varphi(\xi) \sin \frac{k\pi}{l} \xi d\xi, \quad D_k(\tau) = \frac{2}{l} \int_0^l f(\xi, \tau) \sin \frac{k\pi}{l} \xi d\xi.$$

□

附加题 2. 利用傅里叶变换求解

$$\begin{cases} u_{tt} + 2u_t = u_{xx} - u & t > 0, -\infty < x < +\infty, \\ u(x, 0) = 0, u_t(x, 0) = x & -\infty < x < +\infty. \end{cases}$$

解. 对 Cauchy 问题关于 x 作 Fourier 变换得

$$\begin{cases} \hat{u}_{tt} + 2\hat{u}_t = (i\lambda)^2 \hat{u} - \hat{u} = -(\lambda^2 + 1) \hat{u} & t > 0, \lambda \in \mathbb{R}, \\ \hat{u}(\lambda, 0) = 0, \hat{u}_t(\lambda, 0) = \hat{\varphi}(\lambda) & \lambda \in \mathbb{R}, \end{cases}$$

其中 $\varphi(x) = x$. 关于 t 的微分方程的特征方程为 $s^2 + 2s + \lambda^2 + 1 = 0$, 解得 $s = -1 \pm i\lambda$, 且皆为单根. 因此微分方程通解为

$$\hat{u}(\lambda, t) = C_1(\lambda) e^{-t} \cos(\lambda t) + C_2(\lambda) e^{-t} \sin(\lambda t).$$

由 $\hat{u}(\lambda, 0) = 0, \hat{u}_t(\lambda, 0) = \hat{\varphi}(\lambda)$ 得

$$\begin{cases} C_1(\lambda) = 0, \\ \lambda C_2(\lambda) = \hat{\varphi}(\lambda). \end{cases}$$

即 $C_1(\lambda) = 0, C_2(\lambda) = \frac{\hat{\varphi}(\lambda)}{\lambda}$. 故 $\hat{u}(\lambda, t) = \hat{\varphi}(\lambda) \frac{\sin(\lambda t)}{\lambda} e^{-t}$. 注意到

$$\frac{\sin(\lambda t)}{\lambda} = \frac{e^{i\lambda t} - e^{-i\lambda t}}{i2\lambda} = \frac{1}{2} \int_{-t}^t e^{-i\lambda \xi} d\xi = F[f_t](\lambda)$$

其中

$$f_t(x) = \begin{cases} \frac{1}{2}, & |x| \leq t, \\ 0, & |x| > t. \end{cases}$$

因此有

$$\begin{aligned} F^{-1} \left[\frac{\sin(\lambda t)}{\lambda} \right] (x) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \frac{\sin(\lambda t)}{\lambda} e^{i\lambda x} d\lambda \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} F[f_t](\lambda) e^{i\lambda x} d\lambda \\ &= F^{-1}[F[f_t]](x) \\ &= f_t(x). \end{aligned}$$

故

$$\begin{aligned}
u(x, t) &= F^{-1} \left[\hat{\varphi}(\lambda) \frac{\sin(\lambda t)}{\lambda} e^{-t} \right] = e^{-t} \left(\varphi * F^{-1} \left[\frac{\sin(\lambda t)}{\lambda} \right] \right) (x, t) \\
&= e^{-t} \int_{-\infty}^{+\infty} \varphi(x - \xi) f_t(\xi) d\xi \\
&= e^{-t} \int_{-t}^t \frac{1}{2} (x - \xi) d\xi \\
&= -\frac{1}{4} e^{-t} (\xi - x)^2 \Big|_{\xi=-t}^{\xi=t} \\
&= -\frac{1}{4} e^{-t} [(t - x)^2 - (t + x)^2] \\
&= xte^{-t}.
\end{aligned}$$

□

另解 (由杨沅麟提供). 设 $w(x, t) = xg(t)$ 满足原 Cauchy 问题, 即

$$\begin{cases} xg''(t) + 2xg'(t) = -xg(t) & t > 0, x \in \mathbb{R}, \\ xg(0) = 0, xg'(0) = x & x \in \mathbb{R}. \end{cases}$$

故

$$\begin{cases} g''(t) + 2g'(t) + g(t) = 0 & t > 0, \\ g(0) = 0, g'(0) = 1. \end{cases}$$

相应特征方程为 $s^2 + 2s + 1 = 0$, 得 $s = -1$, 为 2 重根, 故常微分方程通解为 $g(t) = Ae^{-t} + Bte^{-t}$. 由 $g(0) = 0, g'(0) = 1$ 得 $A = 0, B = 1$. 故 $g(t) = te^{-t}$. 令 $v(x, t) = u(x, t) - w(x, t) = u(x, t) - xte^{-t}$, 则有

$$\begin{cases} v_{tt} + 2v_t = v_{xx} - v & t > 0, x \in \mathbb{R}, \\ u(x, 0) = 0, v_t(x, 0) = 0 & x \in \mathbb{R}. \end{cases}$$

关于 x 作 Fourier 变换得

$$\begin{cases} \hat{v}_{tt} + 2\hat{v}_t = -(\lambda^2 + 1)\hat{v} & t > 0, \lambda \in \mathbb{R}, \\ \hat{v}(\lambda, 0) = 0, \hat{v}_t(\lambda, 0) = 0 & \lambda \in \mathbb{R}. \end{cases}$$

故 $\hat{v}(\lambda, t) = 0$. 从而 $v(x, t) = F^{-1}[\hat{v}](x, t) = 0$. 于是 $u(x, t) = v(x, t) + xte^{-t} = xte^{-t}$. □

8 第 8 次作业

1. 求解如下初边值问题

$$\begin{cases} u_t = u_{xx} + e^{-t} \sin 3x, & 0 < x < \pi, t > 0, \\ u(x, 0) = x \sin x, & 0 < x < \pi, \\ u(0, t) = u(\pi, t) = 0, & t > 0. \end{cases}$$

解. 将原初边值问题分解为如下两个初边值问题

$$\begin{cases} u_t = u_{xx}, & 0 < x < \pi, t > 0, \\ u(x, 0) = x \sin x, & 0 < x < \pi, \\ u(0, t) = u(\pi, t) = 0, & t > 0 \end{cases} \quad (8.1)$$

与

$$\begin{cases} u_t = u_{xx} + e^{-t} \sin 3x, & 0 < x < \pi, t > 0 \\ u(x, 0) = 0, & 0 < x < \pi, \\ u(0, t) = u(\pi, t) = 0, & t > 0 \end{cases} \quad (8.2)$$

记 (8.1) 的解为 u_1 , (8.2) 的解为 u_2 , 则 $u = u_1 + u_2$ 是原初边值问题的解.

由分离变量法得

$$u_1(x, t) = \sum_{k=1}^{+\infty} A_k e^{-k^2 t} \sin(kx),$$

其中

$$\begin{aligned} A_k &= \frac{2}{\pi} \int_0^{\pi} u(x, 0) \sin(kx) dx \\ &= \frac{2}{\pi} \int_0^{\pi} x \sin x \sin(kx) dx. \end{aligned}$$

当 $k = 1$ 时,

$$\begin{aligned} A_1 &= \frac{2}{\pi} \int_0^{\pi} x \sin^2 x dx \\ &= \frac{1}{\pi} \int_0^{\pi} x(1 - \cos 2x) dx \\ &= 1 - \frac{1}{2\pi} \int_0^{\pi} x \sin 2x dx \\ &= 1 - \frac{1}{2\pi} x \sin 2x \Big|_0^{\pi} + \frac{1}{2\pi} \int_0^{\pi} \sin 2x dx \\ &= 1 - \frac{1}{4\pi} \cos 2x \Big|_0^{\pi} \\ &= 1. \end{aligned}$$

当 $k \geq 2$ 时, 有

$$\begin{aligned}
A_k &= \frac{1}{\pi} \int_0^\pi x [\cos(k-1)x - \cos(k+1)x] dx \\
&= \frac{1}{\pi} \int_0^\pi x d \left\{ \frac{1}{k-1} \sin[(k-1)x] - \frac{1}{k+1} \sin[(k+1)x] \right\} \\
&= \frac{1}{\pi} x \left\{ \frac{1}{k-1} \sin[(k-1)x] - \frac{1}{k+1} \sin[(k+1)x] \right\} \Big|_0^\pi \\
&\quad - \frac{1}{\pi} \int_0^\pi \left\{ \frac{1}{k-1} \sin[(k-1)x] - \frac{1}{k+1} \sin[(k+1)x] \right\} dx \\
&= \frac{1}{\pi} \left\{ \frac{1}{(k-1)^2} \cos[(k-1)x] - \frac{1}{(k+1)^2} \cos[(k+1)x] \right\} \Big|_0^\pi \\
&= \frac{1}{\pi} \left[\frac{(-1)^{k-1} - 1}{(k-1)^2} - \frac{(-1)^{k+1} - 1}{(k+1)^2} \right] \\
&= \frac{(-1)^{k-1} - 1}{\pi} \frac{4k}{(k-1)^2(k+1)^2} \\
&= \begin{cases} -\frac{8}{\pi} \frac{k}{(k^2-1)^2}, & k \text{ 为偶数}, \\ 0, & k \geq 3 \text{ 且 } k \text{ 为奇数}. \end{cases}
\end{aligned}$$

故

$$u_1(x, t) = e^{-t} \sin x - \frac{8}{\pi} \sum_{k=1}^{+\infty} \frac{2k}{(4k^2-1)^2} e^{-4k^2 t} \sin(2kx).$$

由齐次化原理知

$$u_2(x, t) = \int_0^t w(x, t; \tau) d\tau,$$

其中 $w(x, t; \tau)$ 满足

$$\begin{cases} w_t = w_{xx}, & t > \tau, \quad 0 < x < \pi, \\ w(x, \tau; \tau) = e^{-\tau} \sin 3x, & 0 < x < \pi, \\ w(0, t; \tau) = w(\pi, t; \tau) = 0, & t > \tau. \end{cases}$$

令 $t' = t - \tau$, 则有

$$\begin{cases} w_{t'} = w_{xx}, & t' > 0, \quad 0 < x < \pi, \\ t' = 0 : w = e^{-\tau} \sin 3x, & 0 < x < \pi, \\ x = 0 \text{ 及 } x = \pi : & w = 0, \quad t' > 0. \end{cases}$$

从而由前述结果知

$$w(x, t; \tau) = \sum_{k=1}^{+\infty} D_k(\tau) e^{-k^2 t'} \sin(kx) = \sum_{k=1}^{+\infty} D_k(\tau) e^{-k^2(t-\tau)} \sin(kx),$$

其中

$$\begin{aligned}
D_k(\tau) &= \frac{2}{\pi} \int_0^\pi e^{-\tau} \sin 3x \sin kx dx \\
&= \begin{cases} e^{-\tau}, & k = 3, \\ 0, & k \neq 3. \end{cases}
\end{aligned}$$

故 $w(x, t; \tau) = e^{-\tau} e^{-9(t-\tau)} \sin(3x) = e^{-9t} e^{8\tau} \sin(3x)$, 从而得

$$\begin{aligned} u_2(x, t) &= \int_0^t w(x, t; \tau) d\tau \\ &= e^{-9t} \sin(3x) \int_0^t e^{8\tau} d\tau \\ &= \frac{1}{8} e^{-9t} \sin(3x) e^{8\tau} \Big|_0^t \\ &= \frac{1}{8} \sin(3x) (e^{-t} - e^{-9t}). \end{aligned}$$

综上, 有

$$\begin{aligned} u(x, t) &= u_1(x, t) + u_2(x, t) \\ &= e^{-t} \sin x + \frac{1}{8} \sin(3x) (e^{-t} - e^{-9t}) - \frac{8}{\pi} \sum_{k=1}^{\infty} \frac{2k}{(4k^2 - 1)^2} e^{-4k^2 t} \sin(2kx). \end{aligned}$$

□

2. 求解具有放射性衰变的热传导方程的混合问题

$$\begin{cases} u_t = a^2 u_{xx} + A e^{-\alpha x}, & 0 < x < l, \quad t > 0, \\ u(x, 0) = \phi(x), & 0 < x < l, \\ u(0, t) = u(l, t) = 0, & t > 0, \end{cases}$$

其中 $A, \alpha > 0$ 为常数.

解. 将原初值问题分解为如下两个初边值问题

$$\begin{cases} u_t = a^2 u_{xx}, & 0 < x < l, \quad t > 0, \\ u(x, 0) = \phi(x), & 0 < x < l, \\ u(0, l) = u(l, t) = 0, & t > 0 \end{cases} \quad (8.3)$$

与

$$\begin{cases} u_t = a^2 u_{xx} + A e^{-\alpha x}, & 0 < x < l, \quad t > 0, \\ u(x, 0) = 0, & 0 < x < l, \\ u(0, l) = u(l, t) = 0, & t > 0. \end{cases} \quad (8.4)$$

记 (8.3) 的解为 u_1 , (8.4) 的解为 u_2 , 则 $u = u_1 + u_2$ 为原初边值问题的解.

由分离变量法得

$$u_1 = \sum_{k=1}^{+\infty} A_k e^{-\frac{a^2 k^2 \pi^2}{l^2} t} \sin \frac{k\pi}{l} x,$$

其中 $A_k = \frac{2}{l} \int_0^l \phi(x) \sin \frac{k\pi}{l} x dx$.

由齐次化原理知 $u_2 = \int_0^t w(x, t; \tau) d\tau$, 其中 $w(x, t; \tau)$ 满足

$$\begin{cases} w_t = a^2 w_{xx}, & t > \tau, \quad 0 < x < l, \\ w(x, \tau; \tau) = A e^{-\alpha x}, & 0 < x < l, \\ w(0, t; \tau) = w(l, t; \tau) = 0, & t > \tau. \end{cases}$$

令 $t' = t - \tau$, 则有

$$\begin{cases} w_{t'} = a^2 w_{xx}, & t' > 0, 0 < x < l \\ t' = 0 : w = Ae^{-\alpha x}, & 0 < x < l \\ x = 0 \text{ 及 } x = l : w = 0, & t' > 0. \end{cases}$$

故由上述结果知

$$w(x, t; \tau) = \sum_{k=1}^{+\infty} D_k(\tau) e^{-\frac{a^2 k^2 \pi^2}{l^2} t'} \sin \frac{k\pi}{l} x = \sum_{k=1}^{+\infty} D_k(\tau) e^{-\frac{a^2 k^2 \pi^2}{l^2} (t-\tau)} \sin \frac{k\pi}{l} x,$$

其中

$$\begin{aligned} D_k(\tau) &= \frac{2}{l} \int_0^l A e^{-\alpha x} \sin \frac{k\pi}{l} x dx \\ &= \frac{A}{il} \int_0^l e^{-\alpha x} \left(e^{i \frac{k\pi}{l} x} - e^{-i \frac{k\pi}{l} x} \right) dx \\ &= \frac{A}{il} \int_0^l \left[e^{(-\alpha + i \frac{k\pi}{l})x} - e^{-(\alpha + i \frac{k\pi}{l})x} \right] dx \\ &= \frac{A}{il} \left[\frac{1}{-\alpha + i \frac{k\pi}{l}} e^{(-\alpha + i \frac{k\pi}{l})x} + \frac{1}{\alpha + i \frac{k\pi}{l}} e^{-(\alpha + i \frac{k\pi}{l})x} \right] \Big|_0^l \\ &= \frac{A}{il} \left[\frac{1}{-\alpha + i \frac{k\pi}{l}} (e^{-\alpha l + ik\pi} - 1) + \frac{1}{\alpha + i \frac{k\pi}{l}} (e^{-\alpha l - ik\pi} - 1) \right] \\ &= \frac{A}{il} \left(\frac{1}{\alpha - i \frac{k\pi}{l}} - \frac{1}{\alpha + i \frac{k\pi}{l}} \right) + \frac{A}{il} e^{-\alpha l} \left(\frac{1}{-\alpha + i \frac{k\pi}{l}} e^{ik\pi} + \frac{e^{-ik\pi}}{\alpha + i \frac{k\pi}{l}} \right) \\ &= \frac{A}{il} \frac{2i \frac{k\pi}{l}}{\alpha^2 + \frac{k^2 \pi^2}{l^2}} - \frac{A}{il} e^{-\alpha l} \frac{e^{ik\pi} (\alpha + i \frac{k\pi}{l}) - e^{-ik\pi} (\alpha - i \frac{k\pi}{l})}{\alpha^2 + \frac{k^2 \pi^2}{l^2}} \\ &= \frac{2Ak\pi}{\alpha^2 l^2 + k^2 \pi^2} - \frac{A}{i} e^{-\alpha l} \frac{\alpha (e^{ik\pi} - e^{-ik\pi}) + ik\pi (e^{ik\pi} + e^{-ik\pi})}{\alpha^2 l^2 + k^2 \pi^2} \\ &= \frac{2Ak\pi}{\alpha^2 l^2 + k^2 \pi^2} - \frac{2Ae^{-\alpha l}}{\alpha^2 l^2 + k^2 \pi^2} (\alpha l \sin k\pi + k\pi \cos k\pi) \\ &= [1 - (-1)^k e^{-\alpha l}] \frac{2Ak\pi}{\alpha^2 l^2 + k^2 \pi^2}. \end{aligned}$$

故

$$\begin{aligned} u_2(x, t) &= \sum_{k=1}^{+\infty} \int_0^t D_k(\tau) e^{-\frac{a^2 k^2 \pi^2}{l^2} (t-\tau)} \sin \frac{k\pi}{l} x d\tau \\ &= \sum_{k=1}^{+\infty} [1 - (-1)^k e^{-\alpha l}] \frac{2Ak\pi}{\alpha^2 l^2 + k^2 \pi^2} e^{-\frac{a^2 k^2 \pi^2}{l^2} t} \sin \frac{k\pi x}{l} \int_0^t e^{\frac{a^2 k^2 \pi^2}{l^2} \tau} d\tau \\ &= \sum_{k=1}^{+\infty} [1 - (-1)^k e^{-\alpha l}] \frac{2Ak\pi}{\alpha^2 l^2 + k^2 \pi^2} \frac{l^2}{a^2 k^2 \pi^2} e^{-\frac{a^2 k^2 \pi^2}{l^2} t} \sin \frac{k\pi}{l} x \left(e^{\frac{a^2 k^2 \pi^2}{l^2} t} - 1 \right) \\ &= \frac{2Al^2}{a^2 \pi} \sum_{k=1}^{+\infty} \frac{1 - (-1)^k e^{-\alpha l}}{k (\alpha^2 l^2 + k^2 \pi^2)} \left(1 - e^{-\frac{a^2 k^2 \pi^2}{l^2} t} \right) \sin \frac{k\pi}{l} x. \end{aligned}$$

因此

$$\begin{aligned} u(x, t) &= u_1(x, t) + u_2(x, t) \\ &= \sum_{k=1}^{+\infty} A_k e^{-\frac{a^2 k^2 \pi^2}{l^2} t} \sin \frac{kx}{l} + \frac{2Al^2}{a^2 \pi} \sum_{k=1}^{+\infty} \frac{1 - (-1)^k e^{-\alpha l}}{k(\alpha^2 l^2 + k^2 \pi^2)} \left(1 - e^{-\frac{a^2 k^2 \pi^2}{l^2} t}\right) \sin \frac{k\pi}{l} x, \end{aligned}$$

其中 $A_k = \frac{2}{l} \int_0^l \phi(x) \sin \frac{k\pi}{l} x dx$.

□

注 8.1. 在计算 $D_k(\tau)$ 时, 也可由以下两种方式计算积分 $\int_0^l e^{-\alpha x} \sin \left(\frac{k\pi}{l} x \right) dx$.

(1) 记 $\beta = \frac{k\pi}{l}$, 则由

$$\begin{aligned} \int_0^l e^{-\alpha x} \sin \beta x dx &= -\frac{1}{\beta} \int_0^l e^{-\alpha x} d \cos \beta x \\ &= -\frac{1}{\beta} e^{-\alpha x} \cos \beta x \Big|_0^l - \frac{\alpha}{\beta} \int_0^l e^{-\alpha x} \cos \beta x dx \\ &= \frac{1}{\beta} [1 - (-1)^k e^{-\alpha l}] - \frac{\alpha}{\beta^2} \int_0^l e^{-\alpha x} d \sin \beta x \\ &= \frac{1}{\beta} [1 - (-1)^k e^{-\alpha l}] - \frac{\alpha}{\beta^2} e^{-\alpha x} \sin \beta x \Big|_0^l - \frac{\alpha^2}{\beta^2} \int_0^l e^{-\alpha x} \sin \beta x dx \\ &= \frac{1}{\beta} [1 - (-1)^k e^{-\alpha l}] - \frac{\alpha^2}{\beta^2} \int_0^l e^{-\alpha x} \sin \beta x dx \end{aligned}$$

知

$$\int_0^l e^{-\alpha x} \sin \beta x dx = [1 - (-1)^k e^{-\alpha l}] \frac{\beta}{\alpha^2 + \beta^2} = [1 - (-1)^k e^{-\alpha l}] \frac{k\pi l}{\alpha^2 l^2 + k^2 \pi^2}.$$

(2) 记 $f(x) = e^{-\alpha x}$, $g(x) = \sin \left(\frac{k\pi}{l} x \right)$. 则由

$$\begin{aligned} \int_0^l \{[f'(x)g(x)]' - [f(x)g'(x)]'\} dx &= \int_0^l [f''(x)g(x) - f(x)g''(x)] dx \\ &= \left(\alpha^2 + \frac{k^2 \pi^2}{l^2} \right) \int_0^l e^{-\alpha x} \sin \left(\frac{k\pi}{l} x \right) dx \end{aligned}$$

得

$$\begin{aligned} \int_0^l e^{-\alpha x} \sin \left(\frac{k\pi}{l} x \right) dx &= \frac{l^2}{\alpha^2 l^2 + k^2 \pi^2} [f'(x)g(x) - f(x)g'(x)] \Big|_0^l \\ &= [1 - (-1)^k e^{-\alpha l}] \frac{k\pi l}{\alpha^2 l^2 + k^2 \pi^2}. \end{aligned}$$

3. 用分离变量法求解如下初边值问题

$$\begin{cases} u_t = u_{xx} + hu, & 0 < x < \pi, t > 0, \\ u(x, 0) = x(\pi - x), & 0 < x < \pi, \\ u(0, t) = u(\pi, t) = 0, & t > 0, \end{cases}$$

其中 h 为常数.

解. 令 $u(x, t) = X(x)T(t)$, 代方程有 $X(x)T'(t) = X''(x)T(t) + hX(x)T(t)$, 即

$$\frac{T'(t)}{T(t)} - h = \frac{X''(x)}{X(x)} = -\lambda.$$

故

$$\begin{cases} T'(t) + (\lambda - h)T(t) = 0, \\ X''(x) + \lambda X(x) = 0. \end{cases} \quad (8.5)$$

于是有

$$X(x) = \begin{cases} Ae^{\sqrt{\lambda}x} + Be^{-\sqrt{\lambda}x}, & \lambda < 0, \\ A + Bx, & \lambda = 0, \\ A \cos \sqrt{\lambda}x + B \sin \sqrt{\lambda}x, & \lambda > 0. \end{cases}$$

由边界条件 $u(0, t) = u(\pi, t) = 0$ 得 $X(0)T(t) = X(\pi)T(t) = 0$, 即 $X(0) = X(\pi) = 0$. 易知 $\lambda \leq 0$ 时, $X(x) = 0$. 当 $\lambda > 0$ 时, 有

$$\begin{cases} A = 0, \\ B \sin(\sqrt{\lambda}\pi) = 0. \end{cases}$$

为使 $B \neq 0$, 需 $\sin(\sqrt{\lambda}\pi) = 0$, 即 $\sqrt{\lambda}\pi = k\pi, k = 1, 2, \dots$ 故 $\lambda = \lambda_k = k^2, k = 1, 2, \dots$ 相应地 $X_k = B_k \sin kx, k = 1, 2, \dots$ 于是由 (8.5) 知 $T_k(t) = C_k e^{(h-\lambda_k)t} = C_k e^{(h-k^2)t}$. 于是得级数解为

$$u(x, t) = \sum_{k=1}^{+\infty} A_k e^{(h-k^2)t} \sin(kx),$$

其中

$$\begin{aligned} A_k &= \frac{2}{\pi} \int_0^\pi x(\pi - x) \sin(kx) dx \\ &= -\frac{2}{k\pi} \int_0^\pi x(\pi - x) d \cos(kx) \\ &= \frac{2}{k\pi} \int_0^\pi \cos(kx)(\pi - 2x) dx \\ &= \frac{2}{k^2\pi} \int_0^\pi (\pi - 2x) d \sin(kx) \\ &= \frac{2}{k^2\pi} (\pi - 2x) \sin(kx) \Big|_0^\pi + \frac{4}{k^2\pi} \int_0^\pi \sin(kx) dx \\ &= -\frac{4}{k^3\pi} \cos(kx) \Big|_0^\pi \\ &= [1 - (-1)^k] \frac{4}{k^3\pi} \\ &= \begin{cases} \frac{8}{k^3\pi}, & k \text{ 为奇数}, \\ 0, & k \text{ 为偶数}. \end{cases} \end{aligned}$$

故

$$u(x, t) = \frac{8}{\pi} \sum_{k=0}^{+\infty} \frac{1}{(2k+1)^3} e^{[h-(2k+1)^2]t} \sin[(2k+1)x].$$

□

9 第 9 次作业

1. 证明方程

$$\frac{\partial}{\partial x} \left[\left(1 - \frac{x}{h}\right)^2 \frac{\partial u}{\partial x} \right] = \frac{1}{a^2} \left(1 - \frac{x}{h}\right)^2 \frac{\partial^2 u}{\partial t^2} \quad (h > 0, \text{ 常数})$$

的通解可以写成

$$u = \frac{F(x - at) + G(x + at)}{h - x},$$

其中 F, G 为任意的具有二阶连续导数的单变量函数, 并由此求它满足初始条件:

$$t = 0 : u = \varphi(x), \quad \frac{\partial u}{\partial t} = \psi(x)$$

的初值问题的解.

证. 由题设知 $\frac{\partial}{\partial x} [(h - x)^2 u_x] = \frac{1}{a^2} (h - x)^2 u_{tt}$, 即 $(h - x)^2 u_{xx} - 2(h - x)u_x = \frac{1}{a^2} (h - x)^2 u_{tt}$.
故当 $x \neq h$ 时, 有

$$(h - x)u_{xx} - 2u_x = \frac{1}{a^2} (h - x)u_{tt}.$$

令 $v(x, t) = (h - x)u(x, t)$. 则有

$$v_{xx} = -2u_x + (h - x)u_{xx} = \frac{1}{a^2} (h - x)u_{tt} = \frac{1}{a^2} v_{tt},$$

即 $v_{tt} = a^2 v_{xx}$. 于是由行波法知 $v(x, t) = F(x - at) + G(x + at)$, 其中 $F, G \in C^2(\mathbb{R})$. 因此

$$u(x, t) = \frac{F(x - at) + G(x + at)}{h - x}.$$

由初始条件知

$$\begin{cases} F(x) + G(x) = (h - x)\varphi(x), \\ a[-F'(x) + G'(x)] = (h - x)\psi(x), \end{cases}$$

即

$$\begin{cases} F(x) + G(x) = (h - x)\varphi(x), \\ -F(x) + G(x) = \frac{1}{a} \int_0^x (h - \xi)\psi(\xi) d\xi + C. \end{cases}$$

从而得

$$\begin{cases} F(x) = \frac{1}{2}(h - x)\varphi(x) - \frac{1}{2a} \int_0^x (h - \xi)\psi(\xi) d\xi - \frac{1}{2C}, \\ G(x) = \frac{1}{2}(h - x)\varphi(x) + \frac{1}{2a} \int_0^x (h - \xi)\psi(\xi) d\xi + \frac{1}{2C}. \end{cases}$$

于是有

$$\begin{aligned} & u(x, t) \\ &= \frac{1}{h - x} \left[\frac{(h - x + at)\varphi(x - at) + (h - x - at)\varphi(x + at)}{2} \right] + \frac{1}{2a} \int_{x-at}^{x+at} (h - \xi)\psi(\xi) d\xi. \end{aligned}$$

□

1. 问初始条件 $\varphi(x)$ 与 $\psi(x)$ 满足怎样的条件时, 齐次波动方程初值问题的解仅由右传播波组成?

解. 由题意知

$$G(x) = \frac{1}{2}\varphi(x) + \frac{1}{2a} \int_0^x \psi(\xi) d\xi + \frac{1}{2C} \equiv \text{const},$$

即 $G'(x) = 0$. 故

$$a\varphi'(x) + \psi(x) = 0, \quad \forall x \in \mathbb{R}.$$

□

2. 利用传播波法, 求解波动方程的古尔萨 (Goursat) 问题

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} & (-at < x < at, t > 0), \\ u|_{x-at=0} = \varphi(x) & (t > 0), \\ u|_{x+at=0} = \psi(x) & (t > 0), \quad \varphi(0) = \psi(0). \end{cases}$$

解. 由行波法知 $u(x, t) = F(x - at) + G(x + at)$. 由题设条件知

$$\begin{cases} F(0) + G(2x) = \varphi(x), & x \geq 0, \\ F(2x) + G(0) = \psi(x), & x \leq 0. \end{cases}$$

从而有

$$\begin{cases} F(x) = \psi\left(\frac{1}{2}x\right) - G(0), & x \leq 0, \\ G(x) = \varphi\left(\frac{1}{2}x\right) - F(0), & x \geq 0, \\ F(0) + G(0) = \varphi(0) = \psi(0). \end{cases}$$

故

$$u(x, t) = \psi\left(\frac{x - at}{2}\right) + G\left(\frac{x + at}{2}\right) - \varphi(0).$$

□

10 第 10 次作业

5. 求解

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & x > 0, t > 0, \\ u|_{t=0} = \varphi(x), & u_t|_{t=0} = 0, \\ (u_x - ku_t)|_{x=0} = 0, \end{cases}$$

其中 k 为正常数.

解. 将原问题延拓到整个空间 $\mathbb{R}$ 上:

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & x \in \mathbb{R}, t > 0 \\ u(x, 0) = \Phi(x), & u_t(x, 0) = 0, & x \in \mathbb{R}, \\ (u_x - ku_t)|_{x=0} = 0, & t > 0, \end{cases}$$

其中 $\Phi(x) = \varphi(x), \forall x \geq 0$. 故由达朗贝尔公式知

$$u(x, t) = \frac{1}{2}[\Phi(x - at) + \Phi(x + at)].$$

因端点条件知

$$\Phi'(-at) + \Phi'(at) + ka\Phi'(-at) - ka\Phi'(at) = 0, \quad \forall t > 0,$$

即 $\Phi'(-x) = \frac{ka-1}{1+ka}\Phi'(x), \forall x > 0$. 从而对 $\forall x < 0$ 有

$$\begin{aligned} \Phi(x) &= \int_0^x \Phi'(\xi) d\xi + \Phi(0) \\ &= \frac{ka-1}{1+ka} \int_0^x \Phi'(-\xi) d\xi + \Phi(0) \\ &= \frac{1-ka}{1+ka} \int_0^{-x} \Phi'(\xi) d\xi + \varphi(0) \\ &= \frac{1-ka}{1+ka} [\varphi(-x) - \varphi(0)] + \varphi(0) \\ &= \frac{1-ka}{1+ka} \varphi(-x) + \frac{2ka}{1+ka} \varphi(0) \end{aligned}$$

因此, 当 $x \geq at$ 时, 有

$$u(x, t) = \frac{1}{2}[\varphi(x - at) + \varphi(x + at)].$$

当 $0 \leq x < at$ 时, 有

$$\begin{aligned} u(x, t) &= \frac{1}{2} \left[\frac{1-ka}{1+ka} \varphi(at-x) + \frac{2ka}{1+ka} \varphi(0) + \varphi(x+at) \right] \\ &= \frac{1}{2} \varphi(x+at) + \frac{1-ka}{2(1+ka)} \varphi(at-x) + \frac{ka}{1+ka} \varphi(0). \end{aligned}$$

综上, 有

$$u(x, t) = \begin{cases} \frac{1}{2}[\varphi(x+at) + \varphi(x-at)], & x \geq at, \\ \frac{1}{2} \varphi(x+at) + \frac{1-ka}{2(1+ka)} \varphi(at-x) + \frac{ka}{1+ka} \varphi(0), & 0 \leq x < at. \end{cases}$$

□

8. 求解波动方程的初值问题

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = t \sin x & (t > 0, -\infty < x < \infty), \\ u|_{t=0} = 0, \left. \frac{\partial u}{\partial t} \right|_{t=0} = \sin x & (-\infty < x < \infty). \end{cases}$$

解. 将原初值问题拆分成如下两个初值问题:

$$\begin{cases} u_{tt} - u_{xx} = 0, & t > 0, x \in \mathbb{R}, \\ u(x, 0) = 0, u_t(x, 0) = \sin x, & x \in \mathbb{R} \end{cases} \quad (10.1)$$

与

$$\begin{cases} u_{tt} - u_{xx} = t \sin x, & t > 0, x \in \mathbb{R}, \\ u(x, 0) = 0, u_t(x, 0) = 0, & x \in \mathbb{R}. \end{cases} \quad (10.2)$$

记 (10.1) 的解为 u_1 , (10.2) 的解为 u_2 , 则 $u = u_1 + u_2$ 为原初值问题的解.

由达朗贝尔公式知

$$u_1 = \frac{1}{2} \int_{x-t}^{x+t} \sin \xi d\xi = -\frac{1}{2} \cos \xi \Big|_{x-t}^{x+t} = \frac{1}{2} [\cos(x-t) - \cos(x+t)] = \sin x \sin t.$$

由齐次化原理知 $u_2 = \int_0^t w(x, t; \tau) d\tau$, 其中 $w(x, t; \tau)$ 满足

$$\begin{cases} w_{tt} - w_{xx} = 0, & t > \tau, x \in \mathbb{R}, \\ w(x, \tau; \tau) = 0, w_t(x, \tau; \tau) = \tau \sin x, & x \in \mathbb{R}. \end{cases}$$

故

$$w(x, t; \tau) = \frac{1}{2} \int_{x-(t-\tau)}^{x+(t-\tau)} \tau \sin \xi d\xi = \tau \sin x \sin(t-\tau).$$

从而有

$$\begin{aligned} u_2 &= \int_0^t w(x, t; \tau) d\tau \\ &= \sin x \int_0^t \tau \sin(t-\tau) d\tau \\ &= \sin x \int_0^t \tau d \cos(t-\tau) \\ &= \sin x [\tau \cos(t-\tau)]_{\tau=0}^{\tau=t} - \sin x \int_0^t \cos(t-\tau) d\tau \\ &= t \sin x + \sin x \sin(t-\tau) \Big|_{\tau=0}^{\tau=t} \\ &= t \sin x - \sin x \sin t \end{aligned}$$

于是有 $u = u_1 + u_2 = t \sin x$.

□

10. 证明: 在柯西问题

$$\begin{cases} u_{tt} - a^2 u_{xx} = f(x, t) & (t > 0, -\infty < x < \infty), \\ u|_{t=0} = u_t|_{t=0} = 0 & (-\infty < x < \infty) \end{cases}$$

中, 若 $f(x, t)$ 是 x 的奇函数, 则 $u(0, t) \equiv 0$; 若 $f(x, t)$ 是 x 的偶函数, 则 $u_x(0, t) \equiv 0$.

证. 由齐次化原理知

$$u(x, t) = \frac{1}{2a} \int_0^t \int_{x-a(t-\tau)}^{x+a(t-\tau)} f(\xi, \tau) d\xi d\tau.$$

若 $f(x, t)$ 关于 x 是奇函数, 则

$$u(0, t) = \frac{1}{2a} \int_0^t \int_{-a(t-\tau)}^{a(t-\tau)} f(\xi, \tau) d\xi d\tau = \frac{1}{2a} \int_0^t 0 d\tau = 0.$$

若 $f(x, t)$ 关于 x 是偶函数, 由

$$\begin{aligned} u_x(x, t) &= \frac{1}{2a} \int_0^t \frac{\partial}{\partial x} \int_{x-a(t-\tau)}^{x+a(t-\tau)} f(\xi, \tau) d\xi d\tau \\ &= \frac{1}{2a} \int_0^t [f(x+a(t-\tau), \tau) - f(x-a(t-\tau), \tau)] d\tau \end{aligned}$$

得

$$\begin{aligned} u_x(0, t) &= \frac{1}{2a} \int_0^t [f(a(t-\tau), \tau) - f(-a(t-\tau), \tau)] d\tau \\ &= \frac{1}{2a} \int_0^t 0 d\tau \\ &= 0. \end{aligned}$$

结论得证. □

1. 用分离变量法求下列问题的解:

$$\begin{aligned} (1) \quad & \begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} & (t > 0, 0 < x < l), \\ u|_{t=0} = \sin \frac{3\pi x}{l}, \quad \frac{\partial u}{\partial t} \Big|_{t=0} = x(l-x) & (0 < x < l), \\ u(0, t) = u(l, t) = 0 & (t > 0); \end{cases} \\ (2) \quad & \begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = 0 & (t > 0, 0 < x < l), \\ u(0, t) = 0, \quad \frac{\partial u}{\partial x}(l, t) = 0 & (t > 0), \\ u(x, 0) = \frac{h}{l}x, \quad \frac{\partial u}{\partial t}(x, 0) = 0 & (0 < x < l). \end{cases} \end{aligned}$$

解. (1) 令 $u(x, t) = X(x)T(t)$. 代入方程得 $X(x)T''(t) = a^2 X''(x)T(t)$, 即 $\frac{T''(t)}{a^2 T(t)} = \frac{X''(x)}{X(x)} = -\lambda$. 故

$$\begin{cases} T''(t) + a^2 \lambda T(t) = 0 \\ X''(x) + \lambda X(x) = 0 \end{cases}$$

解关于 X 的方程得

$$X(x) = \begin{cases} Ae^{\sqrt{-\lambda}x} + Be^{-\sqrt{-\lambda}x}, & \lambda < 0, \\ A + Bx, & \lambda = 0, \\ A \cos \sqrt{\lambda}x + B \sin \sqrt{\lambda}x, & \lambda > 0. \end{cases}$$

由边界条件得 $X(0)T(t) = X(l)T(t) = 0$, 即 $X(0) = X(l) = 0$. 易知当 $\lambda \leq 0$ 时, $X = 0$. 当 $\lambda > 0$ 时, 有 $A = 0, B \sin \sqrt{\lambda}l = 0$. 为使 $B \neq 0$, 需 $\sin \sqrt{\lambda}l = 0$, 即 $\sqrt{\lambda}l = k\pi, k = 1, 2, \dots$ 故

$$\lambda = \lambda_k = \left(\frac{k\pi}{l}\right)^2, \quad k = 1, 2, \dots$$

相应地,

$$X_k(x) = C_k \sin\left(\frac{k\pi}{l}x\right), \quad k = 1, 2, \dots$$

求解关于 T 的方程得

$$T_k = A_k \cos a\sqrt{\lambda_k}t + B_k \sin a\sqrt{\lambda_k}t = A_k \cos\left(\frac{k\pi a}{l}t\right) + B_k \sin\left(\frac{k\pi a}{l}t\right).$$

于是得原问题级数解为

$$u(x, t) = \sum_{k=1}^{+\infty} \left[A_k \cos\left(\frac{k\pi a}{l}t\right) + B_k \sin\left(\frac{k\pi a}{l}t\right) \right] \sin \frac{k\pi}{l}x,$$

其中

$$\begin{aligned} A_k &= \frac{2}{l} \int_0^l \sin \frac{3\pi}{l}x \sin \frac{k\pi}{l}x dx = \begin{cases} 1, & k = 3, \\ 0, & k \neq 3, \end{cases} \\ B_k &= \frac{2}{k\pi a} \int_0^l x(l-x) \sin \frac{k\pi}{l}x dx \\ &= -\frac{2l}{k^2\pi^2 a} \int_0^l x(l-x) d \cos \frac{k\pi}{l}x \\ &= \frac{2l}{k^2\pi^2 a} \int_0^l (l-2x) \cos \frac{k\pi}{l}x dx \\ &= \frac{2l^2}{k^3\pi^3 a} \int_0^l (l-2x) d \sin \frac{k\pi}{l}x \\ &= \frac{2l^2}{k^3\pi^3 a} (l-2x) \sin \frac{k\pi}{l}x \Big|_0^l + \frac{4l^2}{k^3\pi^3 a} \int_0^l \sin \frac{k\pi}{l}x dx \\ &= -\frac{4l^3}{k^4\pi^4 a} \cos \frac{k\pi}{l}x \Big|_0^l \\ &= [1 - (-1)^k] \frac{4l^3}{k^4\pi^4 a}. \\ &= \begin{cases} \frac{8l^3}{k^4\pi^4 a}, & k \text{ 为奇数}, \\ 0, & k \text{ 为偶数}. \end{cases} \end{aligned}$$

故

$$u(x, t) = \cos\left(\frac{3\pi a}{l}t\right) \sin \frac{3\pi}{l}x + \frac{8l^3}{\pi^4 a} \sum_{k=0}^{+\infty} \frac{1}{(2k+1)^4} \sin\left(\frac{2k+1}{l}\pi at\right) \sin\left(\frac{2k+1}{l}\pi x\right).$$

(2) 令 $u(x, t) = X(x)T(t)$. 代入方程得 $X(x)T''(t) = a^2 X''(x)T(t)$, 即 $\frac{T''(t)}{a^2 T(t)} = \frac{X''(x)}{X(x)} = -\lambda$. 故

$$\begin{cases} T''(t) + a^2 \lambda T(t) = 0 \\ X''(x) + \lambda X(x) = 0 \end{cases}$$

解关于 X 的方程得

$$X(x) = \begin{cases} Ae^{\sqrt{-\lambda}x} + Be^{-\sqrt{-\lambda}x}, & \lambda < 0, \\ A + Bx, & \lambda = 0, \\ A \cos \sqrt{\lambda}x + B \sin \sqrt{\lambda}x, & \lambda > 0. \end{cases}$$

由边界条件得 $X(0)T(t) = X'(l)T(t) = 0$, 即 $X(0) = X'(l) = 0$. 易知当 $\lambda \leq 0$ 时, $X = 0$. 当 $\lambda > 0$ 时, 有 $A = 0$, $B\sqrt{\lambda} \cos \sqrt{\lambda}l = 0$. 为使 $B \neq 0$, 需 $\cos \sqrt{\lambda}l = 0$, 即 $\sqrt{\lambda}l = \frac{2k+1}{2}\pi, k = 0, 1, 2, \dots$ 故

$$\lambda = \lambda_k = \frac{(2k+1)^2}{4l^2}\pi^2, \quad k = 0, 1, 2, \dots$$

相应地,

$$X_k(x) = C_k \sin\left(\frac{2k+1}{2l}\pi x\right), \quad k = 0, 1, 2, \dots$$

求解关于 T 的方程得

$$T_k = A_k \cos a\sqrt{\lambda_k}t + B_k \sin a\sqrt{\lambda_k}t = A_k \cos\left(\frac{2k+1}{2l}\pi at\right) + B_k \sin\left(\frac{2k+1}{2l}\pi at\right).$$

于是得原问题级数解为

$$u(x, t) = \sum_{k=0}^{+\infty} \left[A_k \cos\left(\frac{2k+1}{2l}\pi at\right) + B_k \sin\left(\frac{2k+1}{2l}\pi at\right) \right] \sin\left(\frac{2k+1}{2l}\pi x\right).$$

直接计算可得

$$u_t(x, t) = \sum_{k=0}^{+\infty} \frac{2k+1}{2l}\pi a \left[-A_k \sin\left(\frac{2k+1}{2l}\pi at\right) + B_k \cos\left(\frac{2k+1}{2l}\pi at\right) \right] \sin\left(\frac{2k+1}{2l}\pi x\right).$$

于是由初始条件以及 $\left\{ \sin \left(\frac{2k+1}{2l} \pi x \right) \right\}_{k=0}^{+\infty}$ 关于 $L^2(0, l)$ 内积是正交系知

$$\begin{aligned}
 A_k &= \left[\int_0^l \sin^2 \left(\frac{2k+1}{2l} \pi x \right) dx \right]^{-1} \int_0^l \frac{h}{l} x \sin \left(\frac{2k+1}{2l} \pi x \right) dx \\
 &= -\frac{2h}{l^2} \frac{2l}{(2k+1)\pi} \int_0^l x d \cos \left(\frac{2k+1}{2l} \pi x \right) \\
 &= -\frac{4h}{(2k+1)\pi l} x \cos \left(\frac{2k+1}{2l} \pi x \right) \Big|_0^l + \frac{4h}{(2k+1)\pi l} \int_0^l \cos \left(\frac{2k+1}{2l} \pi x \right) dx \\
 &= \frac{8h}{(2k+1)^2 \pi^2} \sin \left(\frac{2k+1}{2l} \pi x \right) \Big|_0^l \\
 &= (-1)^k \frac{8h}{(2k+1)^2 \pi^2}
 \end{aligned}$$

以及 $B_k = 0$. 故

$$u(x, t) = \frac{8h}{\pi^2} \sum_{k=0}^{+\infty} (-1)^k \frac{1}{(2k+1)^2} \cos \left(\frac{2k+1}{2l} \pi a t \right) \sin \left(\frac{2k+1}{2l} \pi x \right).$$

□

2. 设弹簧一端固定, 一端在外力作用下作周期振动, 此时定解问题归结为

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} & (t > 0, 0 < x < l), \\ u(0, t) = 0, u(l, t) = A \sin^2 \omega t & (t > 0), \\ u(x, 0) = \frac{\partial u}{\partial t}(x, 0) = 0 & (0 < x < l). \end{cases}$$

求解此问题.

解. 令 $v(x, t) = u(x, t) - \frac{x}{l} A \sin^2 \omega t$, 则有 $v(0, t) = u(0, t) = 0, v(l, t) = u(l, t) - A \sin^2 \omega t = 0, v(x, 0) = u(x, 0) = 0, v_t(x, 0) = \left[u_t(x, t) - \frac{\omega A}{l} x \sin(2\omega t) \right] \Big|_{t=0} = u_t(x, 0) = 0,$

$$v_{tt} - a^2 v_{xx} = u_{tt} - a^2 u_{xx} - \left[\frac{\omega A}{l} x \sin(2\omega t) \right]' = -\frac{2\omega^2 A}{l} x \cos(2\omega t).$$

综上 v 满足如下初边值问题

$$\begin{cases} v_{tt} - a^2 v_{xx} = -\frac{2\omega^2 A}{l} x \cos(2\omega t), & 0 < x < l, t > 0, \\ v(x, 0) = v_t(x, 0) = 0, & 0 < x < l, \\ v(0, t) = v(l, t) = 0, & t > 0. \end{cases}$$

由齐次化原理知 $v(x, t) = \int_0^t w(x, t; \tau) d\tau$, 其中 $w(x, t; \tau)$ 满足

$$\begin{cases} w_{tt} - a^2 w_{xx} = 0, & 0 < x < l, t > \tau, \\ w(x, \tau; \tau) = 0, w_t(x, \tau; \tau) = -\frac{2\omega^2 A}{l} x \cos(2\omega \tau), & 0 < x < l, \\ w(0, t; \tau) = w(l, t; \tau) = 0, & t > \tau. \end{cases}$$

故由习题 1(1) 知

$$w(x, t; \tau) = \sum_{k=1}^{+\infty} B_k(\tau) \sin \left[\frac{k\pi a}{l}(t - \tau) \right] \sin \frac{k\pi}{l} x,$$

其中

$$\begin{aligned} B_k(\tau) &= -\frac{2}{k\pi a} \frac{2\omega^2 A}{l} \cos(2\omega\tau) \int_0^l x \sin \frac{k\pi}{l} x dx \\ &= \frac{4\omega^2 A}{k^2 \pi^2 a} \cos(2\omega\tau) \int_0^l x d \cos \frac{k\pi}{l} x \\ &= \frac{4\omega^2 A}{k^2 \pi^2 a} \cos(2\omega\tau) \left(x \cos \frac{k\pi}{l} x \Big|_0^l - \int_0^l \cos \frac{k\pi}{l} x dx \right) \\ &= \frac{4\omega^2 A}{k^2 \pi^2 a} \cos(2\omega\tau) \left[(-1)^k l - \frac{l}{k\pi} \sin \frac{k\pi}{l} x \Big|_0^l \right] \\ &= (-1)^k \frac{4\omega^2 A l}{k^2 \pi^2 a} \cos(2\omega\tau) \end{aligned}$$

又因为

$$\begin{aligned} & \int_0^t \cos(2\omega\tau) \sin \left[\frac{k\pi a}{l}(t - \tau) \right] d\tau \\ &= \frac{1}{2} \int_0^t \left\{ \sin \left[2\omega\tau + \frac{k\pi a}{l}(t - \tau) \right] - \sin \left[2\omega\tau - \frac{k\pi a}{l}(t - \tau) \right] \right\} d\tau \\ &= \frac{1}{2} \left\{ \left[-\frac{1}{2\omega - \frac{k\pi a}{l}} \cos \left[2\omega\tau + \frac{k\pi a}{l}(t - \tau) \right] + \frac{1}{2\omega + \frac{k\pi a}{l}} \cos \left[2\omega\tau - \frac{k\pi a}{l}(t - \tau) \right] \right] \right\} \Big|_0^t \\ &= \frac{1}{2} \left[\frac{l}{2\omega l - k\pi a} \left(\cos \frac{k\pi a}{l} t - \cos 2\omega t \right) + \frac{l}{2\omega l + k\pi a} \left(\cos 2\omega t - \cos \frac{k\pi a}{l} t \right) \right] \\ &= \frac{k\pi a l}{4\omega^2 l^2 - k^2 \pi^2 a^2} \left(\cos \frac{k\pi a}{l} t - \cos 2\omega t \right), \end{aligned}$$

故

$$\begin{aligned} v(x, t) &= \int_0^t w(x, t; \tau) d\tau \\ &= \sum_{k=1}^{+\infty} (-1)^k \frac{4\omega^2 A l}{k^2 \pi^2 a} \int_0^t \cos 2\omega\tau \sin \left[\frac{k\pi a}{l}(t - \tau) \right] d\tau \sin \frac{k\pi}{l} x \\ &= \frac{4\omega^2 A l^2}{\pi} \sum_{k=1}^{+\infty} \frac{(-1)^k}{k (4\omega^2 l^2 - k^2 \pi^2 a^2)} \left(\cos \frac{k\pi a}{l} t - \cos 2\omega t \right) \sin \frac{k\pi}{l} x. \end{aligned}$$

从而

$$\begin{aligned} u(x, t) &= v(x, t) + \frac{x}{l} A \sin^2 \omega t \\ &= \frac{A}{l} x \sin^2 \omega t + \frac{4\omega^2 A l^2}{\pi} \sum_{k=1}^{+\infty} \frac{(-1)^k}{k (4\omega^2 l^2 - k^2 \pi^2 a^2)} \left(\cos \frac{k\pi a}{l} t - \cos 2\omega t \right) \sin \frac{k\pi}{l} x. \end{aligned}$$

□

3. 求弦振动方程

$$u_{tt} - a^2 u_{xx} = 0, \quad 0 < x < l, \quad t > 0$$

满足以下定解条件的解:

$$\begin{aligned} (1) \quad & u|_{x=0} = u_x|_{x=l} = 0, \\ & u|_{t=0} = \sin \frac{3}{2l} \pi x, \quad u_t|_{t=0} = \sin \frac{5}{2l} \pi x; \\ (2) \quad & u_x|_{x=0} = u_x|_{x=l} = 0, \\ & u|_{t=0} = x, \quad u_t|_{t=0} = 0. \end{aligned}$$

解. (1) 由习题 1(2) 知

$$u(x, t) = \sum_{k=1}^{+\infty} \left[A_k \cos \left(\frac{2k+1}{2l} \pi a t \right) + B_k \sin \frac{2k+1}{2l} \pi a t \right] \sin \left(\frac{2k+1}{2l} \pi x \right),$$

其中

$$\begin{aligned} A_k &= \frac{2}{l} \int_0^l \sin \left(\frac{3}{2l} \pi x \right) \sin \left(\frac{2k+1}{2l} \pi x \right) dx = \begin{cases} 1, & k=1, \\ 0, & k \neq 1, \end{cases} \\ B_k &= \frac{4}{(2k+1)\pi a} \int_0^l \sin \left(\frac{5}{2l} \pi x \right) \sin \left(\frac{2k+1}{2l} \pi x \right) dx = \begin{cases} \frac{2l}{5\pi a}, & k=2, \\ 0, & k \neq 2. \end{cases} \end{aligned}$$

故

$$u(x, t) = \cos \left(\frac{3\pi a}{2l} t \right) \sin \left(\frac{3\pi}{2l} x \right) + \frac{2l}{5\pi a} \sin \left(\frac{5\pi a}{2l} t \right) \sin \left(\frac{5\pi}{2l} x \right).$$

(2) 令 $u(x, t) = X(x)T(t)$. 代入方程得 $X(x)T''(t) = a^2 X''(x)T(t)$, 即 $\frac{T''(t)}{a^2 T(t)} = \frac{X''(x)}{X(x)} = -\lambda$. 故

$$\begin{cases} T''(t) + a^2 \lambda T(t) = 0, \\ X''(x) + \lambda X(x) = 0. \end{cases}$$

解关于 X 的方程得

$$X(x) = \begin{cases} Ae^{\sqrt{-\lambda}x} + Be^{-\sqrt{-\lambda}x}, & \lambda < 0, \\ A + Bx, & \lambda = 0, \\ A \cos \sqrt{\lambda}x + B \sin \sqrt{\lambda}x, & \lambda > 0. \end{cases}$$

由边界条件得 $X'(0)T(t) = X'(l)T(t) = 0$, 即 $X'(0) = X'(l) = 0$. 易知当 $\lambda < 0$ 时, $X = 0$. 当 $\lambda = \lambda_0 = 0$ 时, $B = 0$, 故相应地 $X_0(x) = C_0$. 求解关于 T 的方程得

$$T_0 = A_0 + B_0 t.$$

当 $\lambda > 0$ 时, 有 $B = 0$, $-\sqrt{\lambda}A \sin \sqrt{\lambda}l = 0$. 为使 $A \neq 0$, 需 $\sin \sqrt{\lambda}l = 0$, 即 $\sqrt{\lambda}l = k\pi$, $k = 1, 2, \dots$ 故

$$\lambda = \lambda_k = \left(\frac{k\pi}{l} \right)^2, \quad k = 1, 2, \dots$$

相应地,

$$X_k = C_k \cos \frac{k\pi}{l} x, \quad k = 1, 2, \dots$$

求解关于 T 的方程得

$$T_k = A_k \cos a\sqrt{\lambda_k}t + B_k \sin a\sqrt{\lambda_k}t = A_k \cos \left(\frac{k\pi a}{l}t \right) + B_k \sin \left(\frac{k\pi a}{l}t \right), \quad k = 1, 2, \dots$$

综上得原问题级数解为

$$u(x, t) = A_0 + B_0 t + \sum_{k=1}^{+\infty} \left[A_k \cos \left(\frac{k\pi a}{l}t \right) + B_k \sin \left(\frac{k\pi a}{l}t \right) \right] \cos \frac{k\pi}{l} x.$$

由 $u_t(x, 0) = 0$ 得 $B_k = 0, k = 0, 1, 2, \dots$ 由 $u(x, 0) = x$ 及 $\left\{ \cos \frac{k\pi}{l} x \right\}_{k=0}^{+\infty}$ 是关于 $L^2(0, l)$

的正交系知 $A_0 = \frac{1}{l} \int_0^l x dx = \frac{l}{2}$,

$$\begin{aligned} A_k &= \frac{2}{l} \int_0^l x \cos \frac{k\pi}{l} x dx \\ &= \frac{2}{k\pi} \int_0^l x d \sin \frac{k\pi}{l} x \\ &= -\frac{2}{k\pi} \int_0^l \sin \frac{k\pi}{l} x dx \\ &= \frac{2l}{k^2\pi^2} \cos \frac{k\pi}{l} x \Big|_0^l \\ &= [(-1)^k - 1] \frac{2l}{k^2\pi^2} \\ &= \begin{cases} -\frac{4l}{k^2\pi^2}, & k \text{ 为正奇数,} \\ 0, & k \text{ 为正偶数.} \end{cases} \end{aligned}$$

故

$$u(x, t) = \frac{l}{2} - \frac{4l}{\pi^2} \sum_{k=0}^{+\infty} \frac{1}{(2k+1)^2} \cos \left(\frac{2k+1}{l} \pi a t \right) \cos \left(\frac{2k+1}{l} \pi x \right).$$

□

4. 用分离变量法求解初边值问题:

$$\begin{cases} u_{tt} - a^2 u_{xx} = g, & 0 < x < l, t > 0, \\ u|_{x=0} = u_x|_{x=l} = 0, & t > 0, \\ u|_{t=0} = 0, u_t|_{t=0} = \sin \frac{\pi x}{2l}, & 0 < x < l. \end{cases}$$

其中 g 为常数.

解. 将原问题分解为如下两个初边值问题

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & 0 < x < l, t > 0, \\ u(0, t) = u_x(l, t) = 0, & t > 0, \\ u(x, 0) = 0, u_t(x, 0) = \sin \frac{\pi x}{2l}, & 0 < x < l \end{cases} \quad (10.3)$$

与

$$\begin{cases} u_{tt} - a^2 u_{xx} = g, & 0 < x < l, t > 0, \\ u(0, t) = u_x(l, t) = 0, & t > 0, \\ u(x, 0) = u_t(x, 0) = 0, & 0 < x < l. \end{cases} \quad (10.4)$$

记 (10.3) 的解为 u_1 , (10.4) 的解为 u_2 , 则 $u = u_1 + u_2$ 是原初边值问题的解.

先求 u_1 , 由习题 1(2) 知

$$u_1(x, t) = \sum_{k=0}^{+\infty} \left[A_k \cos \left(\frac{2k+1}{2l} \pi a t \right) + B_k \sin \left(\frac{2k+1}{2l} \pi a t \right) \right] \sin \left(\frac{2k+1}{2l} \pi x \right),$$

其中 $A_k = 0$,

$$\begin{aligned} B_k &= \frac{2l}{(2k+1)\pi a} \frac{2}{l} \int_0^l \sin \frac{\pi x}{2l} \sin \left(\frac{2k+1}{2l} \pi x \right) dx \\ &= \begin{cases} \frac{2l}{\pi a}, & k = 0, \\ 0, & k \neq 0. \end{cases} \end{aligned}$$

故

$$u_1(x, t) = \frac{2l}{\pi a} \sin \frac{\pi a t}{2l} \sin \frac{\pi x}{2l}.$$

由齐次化原理知 $u_2(x, t) = \int_0^t w(x, t; \tau) d\tau$, 其中 $w(x, t; \tau)$ 满足

$$\begin{cases} w_{tt} - a^2 w_{xx} = 0, & 0 < x < l, t > \tau, \\ w(0, t; \tau) = w_x(l, t; \tau) = 0 & t > \tau, \\ w(x, \tau; \tau) = 0, w_t(x, \tau; \tau) = g, & 0 < x < l. \end{cases}$$

故

$$w(x, t; \tau) = \sum_{k=0}^{+\infty} D_k(\tau) \sin \left[\frac{2k+1}{2l} \pi a(t - \tau) \right] \sin \left(\frac{2k+1}{2l} \pi x \right),$$

其中

$$\begin{aligned} D_k(\tau) &= \frac{4}{(2k+1)\pi a} \int_0^l g \sin \left(\frac{2k+1}{2l} \pi x \right) dx \\ &= -\frac{8lg}{(2k+1)^2 \pi^2 a} \cos \left(\frac{2k+1}{2l} \pi x \right) \Big|_0^l \\ &= \frac{8lg}{(2k+1)^2 \pi^2 a}. \end{aligned}$$

由

$$\begin{aligned} & \int_0^t \sin \left[\frac{2k+1}{2l} \pi a(t - \tau) \right] d\tau \\ &= \frac{2l}{(2k+1)\pi a} \cos \left[\frac{2k+1}{2l} \pi a(t - \tau) \right] \Big|_{\tau=0}^{\tau=t} \\ &= \frac{2l}{(2k+1)\pi a} \left[1 - \cos \left(\frac{2k+1}{2l} \pi a t \right) \right] \end{aligned}$$

知

$$u_2(x, t) = \int_0^t w(x, t; \tau) d\tau = \sum_{k=0}^{+\infty} \frac{16l^2 g}{(2k+1)^3 \pi^3 a^2} \left[1 - \cos \left(\frac{2k+1}{2l} \pi a t \right) \right] \sin \left(\frac{2k+1}{2l} \pi x \right)$$

综上, 有

$$\begin{aligned} u(x, t) &= u_1(x, t) + u_2(x, t) \\ &= \frac{2l}{\pi a} \sin \frac{\pi a t}{2l} \sin \frac{\pi x}{2l} + \frac{16l^2 g}{\pi^3 a^2} \sum_{k=0}^{+\infty} \frac{1}{(2k+1)^3} \left[1 - \cos \left(\frac{2k+1}{2l} \pi a t \right) \right] \sin \left(\frac{2k+1}{2l} \pi x \right). \end{aligned}$$

□

5. 用分离变量法求下面问题的解:

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} + b \operatorname{sh} x, & t > 0, 0 < x < l, \\ u|_{x=0} = u|_{x=l} = 0, & t > 0, \\ u|_{t=0} = \frac{\partial u}{\partial t} \Big|_{t=0} = 0, & 0 < x < l, \end{cases}$$

其中 $\operatorname{sh} x = \frac{e^x - e^{-x}}{2}$ 为双曲正弦函数.

解. 由齐次化原理知 $u(x, t) = \int_0^t w(x, t; \tau) d\tau$, 其中 $w(x, t; \tau)$ 满足

$$\begin{cases} w_{tt} - a^2 w_{xx} = 0, & t > \tau, 0 < x < l, \\ w(0, t; \tau) = w(l, t; \tau) = 0, & t > \tau, \\ w(x, \tau; \tau) = 0, w_t(x, \tau; \tau) = b \operatorname{sh} x, & 0 < x < l. \end{cases}$$

由习题 1(1) 知

$$w(x, t; \tau) = \sum_{k=1}^{+\infty} B_k(\tau) \sin \left[\frac{k\pi a}{l} (t - \tau) \right] \sin \left(\frac{k\pi}{l} x \right),$$

其中

$$\begin{aligned}
B_k(\tau) &= \frac{2}{k\pi a} \int_0^l b \operatorname{sh} x \sin\left(\frac{k\pi}{l}x\right) dx \\
&= \frac{b}{k\pi a} \int_0^l \left[e^x \sin\left(\frac{k\pi}{l}x\right) - e^{-x} \sin\left(\frac{k\pi}{l}x\right) \right] dx \\
&= \frac{b}{i2k\pi a} \int_0^l \left[e^x \left(e^{i\frac{k\pi}{l}x} - e^{-i\frac{k\pi}{l}x} \right) - e^{-x} \left(e^{i\frac{k\pi}{l}x} - e^{-i\frac{k\pi}{l}x} \right) \right] dx \\
&= \frac{b}{i2k\pi a} \left[\frac{1}{1+i\frac{k\pi}{l}} e^{(1+i\frac{k\pi}{l})x} - \frac{1}{1-i\frac{k\pi}{l}} e^{(1-i\frac{k\pi}{l})x} - \frac{1}{-1+i\frac{k\pi}{l}} e^{(-1+i\frac{k\pi}{l})x} - \frac{1}{1+i\frac{k\pi}{l}} e^{-(1+i\frac{k\pi}{l})x} \right] \Big|_0^l \\
&= \frac{b}{i2k\pi a} \left[\frac{l}{l+ik\pi} (e^{l+ik\pi} - 1 - e^{-(l+ik\pi)} + 1) - \frac{l}{l-ik\pi} (e^{l-ik\pi} - 1 - e^{-l+ik\pi} + 1) \right] \\
&= \frac{lb}{i2k\pi a} \left[e^l \left(\frac{e^{ik\pi}}{l+ik\pi} - \frac{e^{-ik\pi}}{l-ik\pi} \right) + e^{-l} \left(\frac{e^{ik\pi}}{l-ik\pi} - \frac{e^{-ik\pi}}{l+ik\pi} \right) \right] \\
&= (-1)^k \frac{lb}{i2k\pi a} \frac{-i2k\pi}{l^2 + k^2\pi^2} (e^l - e^{-l}) \\
&= (-1)^{k+1} \frac{2lb}{a(l^2 + k^2\pi^2)} \operatorname{sh} l
\end{aligned}$$

从而由

$$\int_0^t \sin\left[\frac{k\pi a}{l}(t-\tau)\right] d\tau = \frac{l}{k\pi a} \cos\left[\frac{k\pi a}{l}(t-\tau)\right] \Big|_{\tau=0}^{\tau=t} = \frac{l}{k\pi a} \left[1 - \cos\left(\frac{k\pi a}{l}t\right) \right]$$

知

$$\begin{aligned}
u(x, t) &= \int_0^t w(x, t; \tau) d\tau \\
&= \sum_{k=1}^{+\infty} (-1)^{k+1} \frac{2lb}{a(l^2 + k^2\pi^2)} \operatorname{sh} l \frac{l}{k\pi a} \left[1 - \cos\left(\frac{k\pi a}{l}t\right) \right] \sin\left(\frac{k\pi}{l}x\right) \\
&= \frac{2l^2b}{\pi a^2} \operatorname{sh} l \sum_{k=1}^{+\infty} \frac{(-1)^{k+1}}{k(l^2 + k^2\pi^2)} \left[1 - \cos\left(\frac{k\pi a}{l}t\right) \right] \sin\left(\frac{k\pi}{l}x\right).
\end{aligned}$$

□

注 10.1. 在计算 $D_k(\tau)$ 时, 也可由以下两种方式计算积分 $\int_0^l \operatorname{sh} x \sin\left(\frac{k\pi}{l}x\right) dx$.

(1) 记 $\alpha = \frac{k\pi}{l}$, 则有

$$\begin{aligned}
 \int_0^l \operatorname{sh} x \sin \alpha x dx &= -\frac{1}{\alpha} \int_0^l \operatorname{sh} x d \cos \alpha x \\
 &= -\frac{1}{\alpha} \operatorname{sh} x \cos \alpha x \Big|_0^l + \frac{1}{\alpha} \int_0^l \operatorname{ch} x \cos \alpha x dx \\
 &= -\frac{1}{\alpha} (-1)^k \operatorname{sh} l + \frac{1}{\alpha^2} \int_0^l \operatorname{ch} x d \sin \alpha x \\
 &= \frac{1}{\alpha} (-1)^{k+1} \operatorname{sh} l + \frac{1}{\alpha^2} \operatorname{ch} x \sin \alpha x \Big|_0^l - \frac{1}{\alpha^2} \int_0^l \sin \alpha x \operatorname{sh} x dx \\
 &= \frac{1}{\alpha} (-1)^{k+1} \operatorname{sh} l - \frac{1}{\alpha^2} \int_0^l \sin \alpha x \operatorname{sh} x dx,
 \end{aligned}$$

其中 $\operatorname{ch} x = \frac{e^x + e^{-x}}{2}$ 为双曲余弦函数. 故

$$\int_0^l \operatorname{sh} x \sin \left(\frac{k\pi}{l} x \right) dx = (-1)^{k+1} \frac{1}{1 + \frac{l^2}{k^2 \pi^2}} \frac{l}{k\pi} \operatorname{sh} l = (-1)^{k+1} \frac{k\pi l}{l^2 + k^2 \pi^2} \operatorname{sh} l.$$

(2) 记 $f(x) = \operatorname{sh} x$, $g(x) = \sin \left(\frac{k\pi}{l} x \right)$. 则由

$$\begin{aligned}
 \int_0^l \{[f'(x)g(x)]' - [f(x)g'(x)]'\} dx &= \int_0^l [f''(x)g(x) - f(x)g''(x)] dx \\
 &= \left(1 + \frac{k^2 \pi^2}{l^2} \right) \int_0^l \operatorname{sh} x \sin \left(\frac{k\pi}{l} x \right) dx
 \end{aligned}$$

得

$$\begin{aligned}
 \int_0^l \operatorname{sh} x \sin \left(\frac{k\pi}{l} x \right) dx &= \frac{l^2}{l^2 + k^2 \pi^2} [f'(x)g(x) - f(x)g'(x)] \Big|_0^l \\
 &= (-1)^{k+1} \frac{k\pi l}{l^2 + k^2 \pi^2} \operatorname{sh} l.
 \end{aligned}$$

6. 用分离变量法求下面问题的解:

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} + 2b \frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2} & (t > 0, 0 < x < l), \\ u|_{x=0} = u|_{x=l} = 0 & (t > 0), \\ u|_{t=0} = \frac{h}{l} x, \quad \frac{\partial u}{\partial t} \Big|_{t=0} = 0 & (0 < x < l), \end{cases}$$

其中 $b > 0$ 为给定常数.

解. 令 $u(x, t) = X(x)T(t)$. 代入方程得 $X(x)T''(t) + 2bX(x)T'(t) = a^2 X''(x)T(t)$, 从而有

$$\frac{T''(t) + 2bT'(t)}{a^2 T(t)} = \frac{X''(x)}{X(x)} = -\lambda.$$

故

$$\begin{cases} T''(t) + 2bT'(t) + a^2 \lambda T(t) = 0, \\ X''(x) + \lambda X(x) = 0. \end{cases}$$

由边界条件得 $X(0)T(t) = X(l)T(t) = 0$, 即 $X(0) = X(l) = 0$. 解关于 X 的方程得

$$X(x) = \begin{cases} Ae^{\sqrt{-\lambda}x} + Be^{-\sqrt{-\lambda}x}, & \lambda < 0, \\ A + Bx, & \lambda = 0, \\ A \cos \sqrt{\lambda}x + B \sin \sqrt{\lambda}x, & \lambda > 0. \end{cases}$$

当 $\lambda \leq 0$ 时, 易知 $X = 0$. 当 $\lambda > 0$ 时, 有 $A = 0, B \sin \sqrt{\lambda}l = 0$. 为使 $B \neq 0$, 需 $\sin \sqrt{\lambda}l = 0$, 即 $\sqrt{\lambda}l = k\pi, k = 1, 2, \dots$ 故

$$\lambda = \lambda_k = \left(\frac{k\pi}{l}\right)^2, \quad k = 1, 2, \dots$$

相应地,

$$X_k(x) = C_k \sin\left(\frac{k\pi}{l}x\right), \quad k = 1, 2, \dots$$

下面求解 T 的方程, 其特征方程为 $s^2 + 2bs + a^2\lambda = 0$, 即 $(s+b)^2 = b^2 - a^2\lambda$. 故

$$s = \begin{cases} -b \pm \sqrt{b^2 - a^2\lambda}, & \lambda < \frac{b^2}{a^2}, \\ -b, & \lambda = \frac{b^2}{a^2}, \\ -b \pm i\sqrt{a^2\lambda - b^2}, & \lambda > \frac{b^2}{a^2}. \end{cases}$$

因此有

$$T(t) = \begin{cases} Ae^{(-b+\sqrt{b^2-a^2\lambda})t} + Be^{-(b+\sqrt{b^2-a^2\lambda})t}, & \lambda < \frac{b^2}{a^2}, \\ (A+Bt)e^{-bt}, & \lambda = \frac{b^2}{a^2}, \\ \left(A \cos \sqrt{a^2\lambda - b^2}t + B \sin \sqrt{a^2\lambda - b^2}t\right) e^{-bt}, & \lambda > \frac{b^2}{a^2}. \end{cases}$$

记

$$I_1 = \left\{k \in \mathbb{N} : 0 < \frac{k\pi}{l} < \frac{b}{a}\right\}, I_2 = \left\{k \in \mathbb{N} : \frac{k\pi}{l} = \frac{b}{a}\right\}, I_3 = \left\{k \in \mathbb{N} : \frac{k\pi}{l} > \frac{b}{a}\right\},$$

$$\alpha_k = \begin{cases} \sqrt{b^2 - a^2\lambda_k} = \frac{1}{l}\sqrt{b^2l^2 - k^2\pi^2a^2}, & k \in I_1, \\ \sqrt{a^2\lambda_k - b^2} = \frac{1}{l}\sqrt{k^2\pi^2a^2 - b^2l^2}, & k \in I_3. \end{cases}$$

则有

$$T_k(t) = \begin{cases} A_k e^{(-b+\alpha_k)t} + B_k e^{-(b+\alpha_k)t}, & k \in I_1, \\ (A_k + B_k t) e^{-bt}, & k \in I_2, \\ (A_k \cos \alpha_k t + B_k \sin \alpha_k t) e^{-bt}, & k \in I_3. \end{cases}$$

由叠加原理构造级数解

$$\begin{aligned} u(x, t) &= \sum_{k \in I_1} [A_k e^{(-b+\alpha_k)t} + B_k e^{-(b+\alpha_k)t}] \sin\left(\frac{k\pi}{l}x\right) + \sum_{k \in I_2} (A_k + B_k t) e^{-bt} \sin\left(\frac{k\pi}{l}x\right) \\ &\quad + \sum_{k \in I_3} (A_k \cos \alpha_k t + B_k \sin \alpha_k t) e^{-bt} \sin\left(\frac{k\pi}{l}x\right) \end{aligned}$$

由 $u(x, 0) = \frac{h}{l}x$ 得

$$\begin{cases} A_k + B_k = \frac{2}{l} \int_0^l \frac{h}{l} x \sin \frac{k\pi}{l} x dx, & k \in I_1, \\ A_k = \frac{2}{l} \int_0^l \frac{h}{l} x \sin \frac{k\pi}{l} x dx, & k \in I_2 \cup I_3. \end{cases}$$

由

$$\begin{aligned} \int_0^l x \sin \left(\frac{k\pi}{l} x \right) dx &= -\frac{l}{k\pi} \int_0^l x d \cos \left(\frac{k\pi}{l} x \right) \\ &= -\frac{l}{k\pi} x \cos \left(\frac{k\pi}{l} x \right) \Big|_0^l + \frac{l}{k\pi} \int_0^l \cos \left(\frac{k\pi}{l} x \right) dx \\ &= (-1)^{k+1} \frac{l^2}{k\pi} \end{aligned}$$

知

$$\begin{cases} A_k + B_k = (-1)^{k+1} \frac{2h}{k\pi}, & k \in I_1, \\ A_k = (-1)^{k+1} \frac{2h}{k\pi}, & k \in I_2 \cup I_3. \end{cases} \quad (10.5)$$

由 $u_t(x, 0) = 0$ 得

$$\begin{cases} (-b + \alpha_k) A_k - (b + \alpha_k) B_k = 0, & k \in I_1, \\ -bA_k + B_k = 0, & k \in I_2, \\ -bA_k + \alpha_k B_k = 0, & k \in I_3. \end{cases} \quad (10.6)$$

由 (10.5) 与 (10.6) 解得

$$\begin{cases} A_k = (-1)^{k+1} \frac{h(\alpha_k + b)}{\alpha_k k\pi}, B_k = (-1)^{k+1} \frac{h(\alpha_k - b)}{\alpha_k k\pi}, & k \in I_1, \\ A_k = (-1)^{k+1} \frac{2h}{k\pi}, B_k = (-1)^{k+1} \frac{2bh}{k\pi}, & k \in I_2 \\ A_k = (-1)^{k+1} \frac{2h}{k\pi}, B_k = (-1)^{k+1} \frac{2bh}{\alpha_k k\pi}, & k \in I_3. \end{cases}$$

故

$$\begin{aligned} u(x, t) &= \frac{h}{\pi} \sum_{k \in I_1} (-1)^{k+1} \frac{1}{k} \left[\left(1 + \frac{b}{\alpha_k} \right) e^{(-b+\alpha_k)t} + \left(1 - \frac{b}{\alpha_k} \right) e^{-(b+\alpha_k)t} \right] \sin \left(\frac{k\pi}{l} x \right) \\ &\quad + \frac{2h}{\pi} \sum_{k \in I_2} (-1)^{k+1} \frac{1}{k} (1 + bt) e^{-bt} \sin \left(\frac{k\pi}{l} x \right) \\ &\quad + \frac{2h}{\pi} \sum_{k \in I_3} (-1)^{k+1} \frac{1}{k} \left(\cos \alpha_k t + \frac{b}{\alpha_k} \sin \alpha_k t \right) e^{-bt} \sin \left(\frac{k\pi}{l} x \right), \end{aligned}$$

其中

$$\begin{aligned} I_1 &= \left\{ k \in \mathbb{N} : 0 < \frac{k\pi}{l} < \frac{b}{a} \right\}, I_2 = \left\{ k \in \mathbb{N} : \frac{k\pi}{l} = \frac{b}{a} \right\}, I_3 = \left\{ k \in \mathbb{N} : \frac{k\pi}{l} > \frac{b}{a} \right\}, \\ \alpha_k &= \begin{cases} \frac{1}{l} \sqrt{b^2 l^2 - k^2 \pi^2 a^2}, & k \in I_1, \\ \frac{1}{l} \sqrt{k^2 \pi^2 a^2 - b^2 l^2}, & k \in I_3. \end{cases} \end{aligned}$$

□

11 第 11 次作业

1. 对受摩擦力作用且具固定端点的有界弦振动, 满足方程

$$u_{tt} = a^2 u_{xx} - cu_t,$$

其中常数 $c > 0$, 证明其能量是减少的, 并由此证明方程

$$u_{tt} = a^2 u_{xx} - cu_t + f$$

的初边值问题解的唯一性.

证. 设 u_1, u_2 均是

$$\begin{cases} u_{tt} - a^2 u_{xx} + cu_t = f, & t > 0, 0 < x < l, \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & 0 < x < l, \\ u(0, t) = u(l, t) = 0, & t > 0 \end{cases}$$

的解. 令 $v = u_1 - u_2$, 则有

$$\begin{cases} v_{tt} - a^2 v_{xx} + cv_t = 0, & t > 0, 0 < x < l, \\ v(x, 0) = v_t(x, 0) = 0, & 0 < x < l, \\ v(0, t) = v(l, t) = 0, & t > 0. \end{cases}$$

定义能量

$$E(t) = \int_0^l (v_t^2 + a^2 v_x^2) dx, \quad t \geq 0,$$

则有

$$\begin{aligned} E'(t) &= 2 \int_0^l (v_t v_{tt} + a^2 v_x v_{xt}) dx \\ &= 2 \int_0^l v_t (v_{tt} - a^2 v_{xx}) dx + 2a^2 v_x v_t \Big|_0^l \\ &= -2c \int_0^l v_t^2 dx \\ &\leq 0, \end{aligned}$$

其中用到了

$$v_t(0, t) = \frac{\partial}{\partial t}(v(0, t)) = 0, v_t(l, t) = \frac{\partial}{\partial t}(v(l, t)) = 0.$$

故 $E(t)$ 是单调递减函数. 由于

$$E(0) = \int_0^l [v_t^2(x, 0) + a^2 v_x^2(x, 0)] dx = 0,$$

故

$$0 \leq E(t) \leq E(0) = 0,$$

即 $E(t) = 0, t \geq 0$. 从而 $v_t \equiv 0, v_x \equiv 0$. 故 v 恒为常数. 因此

$$v(x, t) = v(x, 0) = 0, \quad t \geq 0, 0 \leq x \leq l.$$

于是在 $[0, l] \times [0, +\infty)$ 上 $u_1 \equiv u_2$, 即原初边值问题的解是唯一的. □